

Abstract Linear Algebra

Prof. Lanier

jlancier8@uchicago.edu

Pick up a handout. Try those warm-up problems.

Next up:

- HW₃ debrief today 4-5pm, Eckhart 207A
- Problem swap solutions are due tonight
- Quizzes 6 & 7 on Friday (finding a basis for a span, change of basis)
- Problem session on Friday, 3-5pm, Ryerson 177.
- HW₄ due Friday / Saturday.
- Midterm exam review on Sunday in Cobb 115, 3:00-4:30pm
- Midterm exam on Monday

Warm-up 1

Using the standard basis, the vector v is given as $\begin{pmatrix} 5 \\ 1 \end{pmatrix}$. Write v using the basis

$b_1 = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$, $b_2 = \begin{pmatrix} -7 \\ 5 \end{pmatrix}$, where these new basis vectors are also given in the standard basis.

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In $\mathbb{F}^n$, to convert from the standard basis
to a new basis $B = \{b_1, \dots, b_n\}$,
apply the matrix

$$\begin{pmatrix} | & | & | \\ b_1 & \dots & b_n \\ | & | & | \end{pmatrix}^{-1}$$

on the left.

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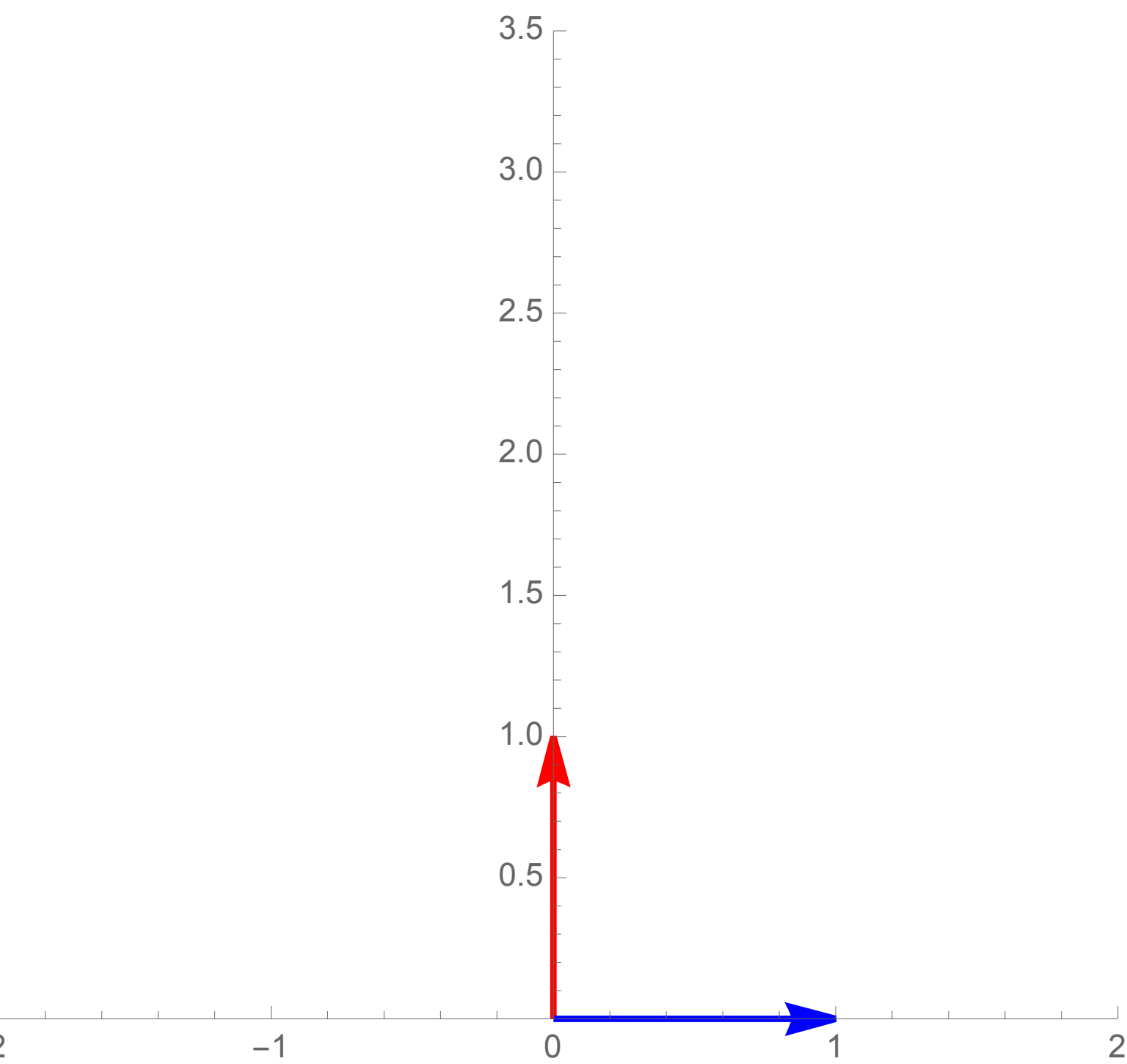
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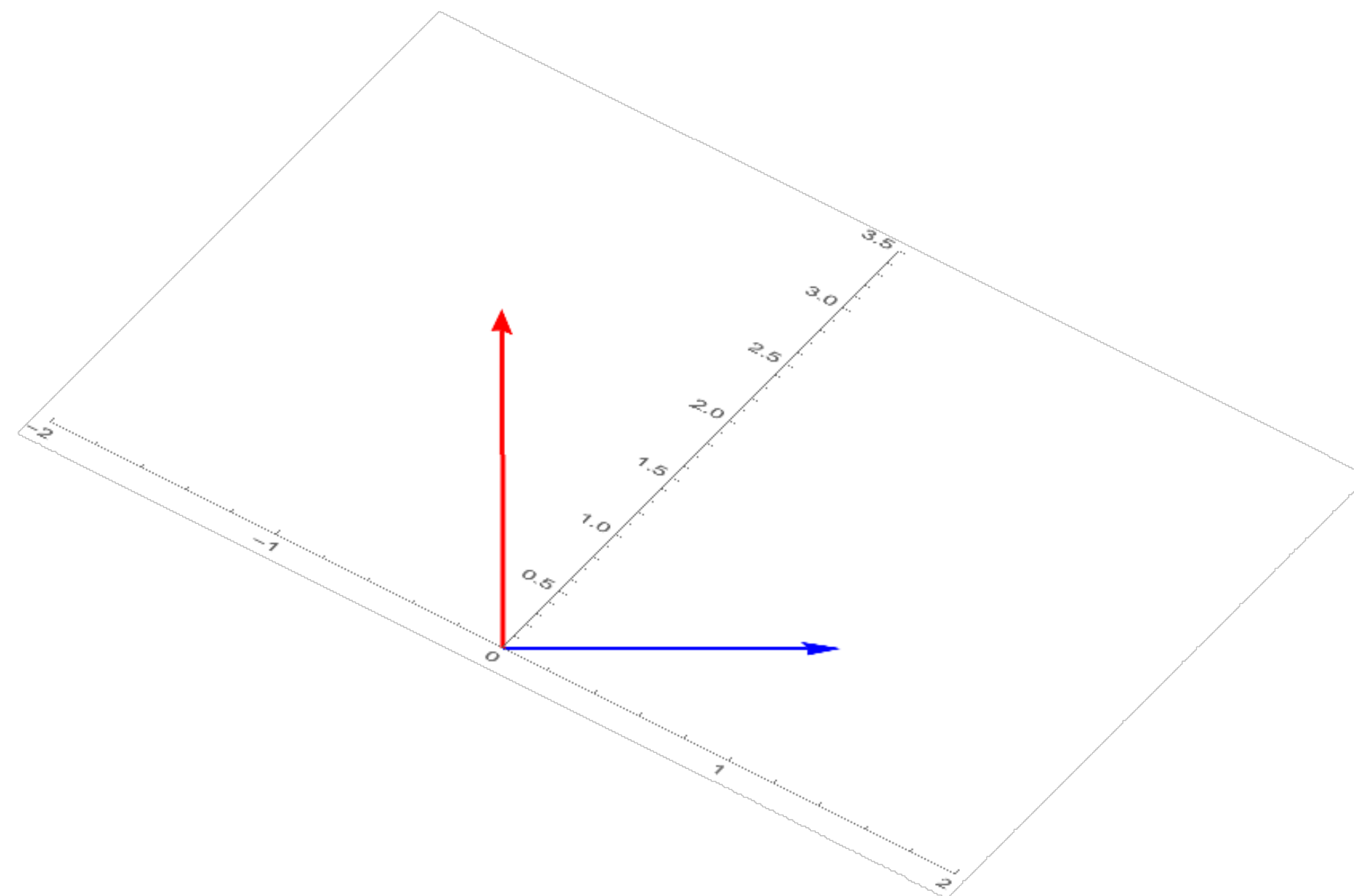
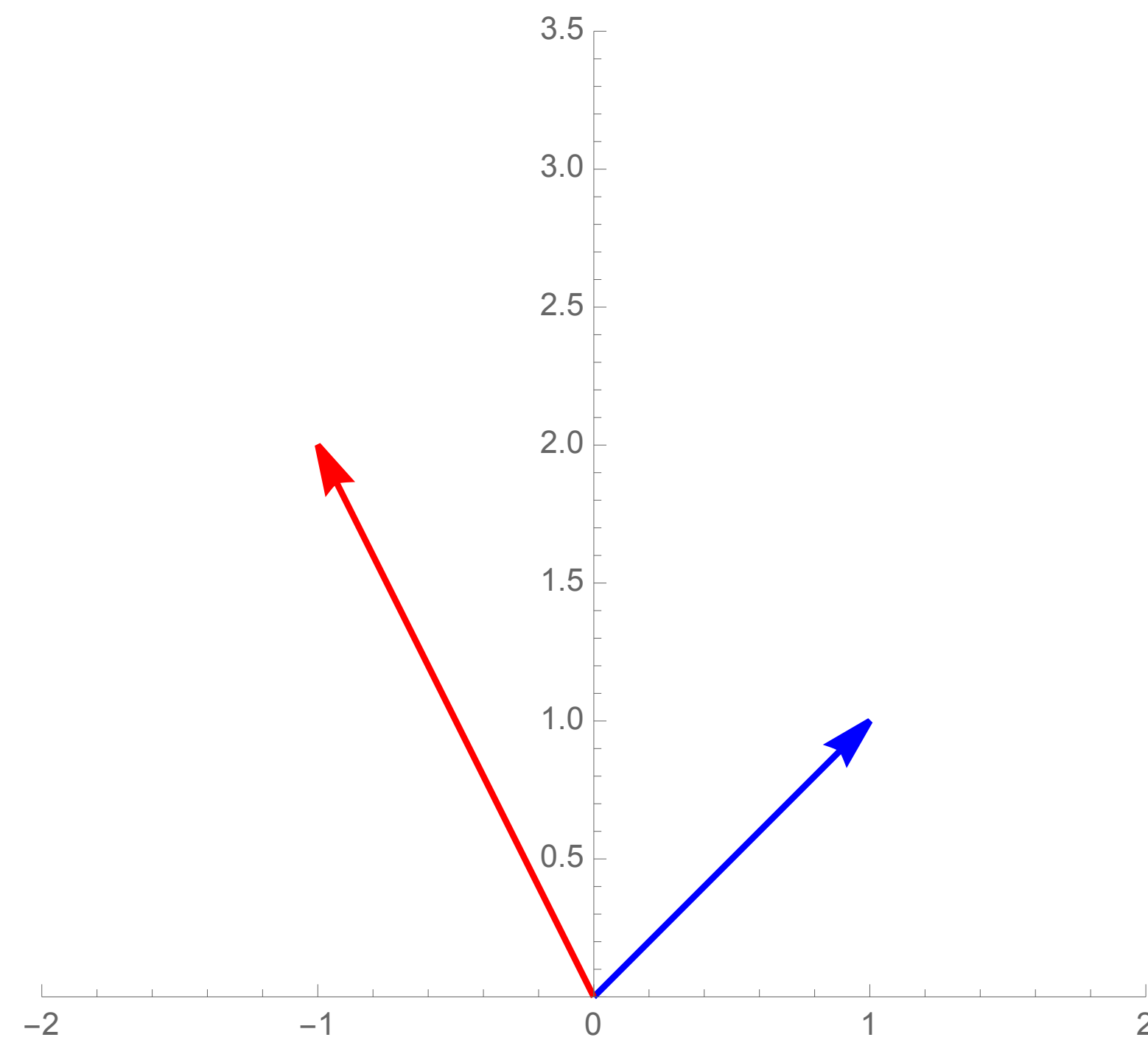
In $\mathbb{F}^n$, to convert from a basis
 $B = \{b_1, \dots, b_n\}$ to the standard basis,
apply the matrix

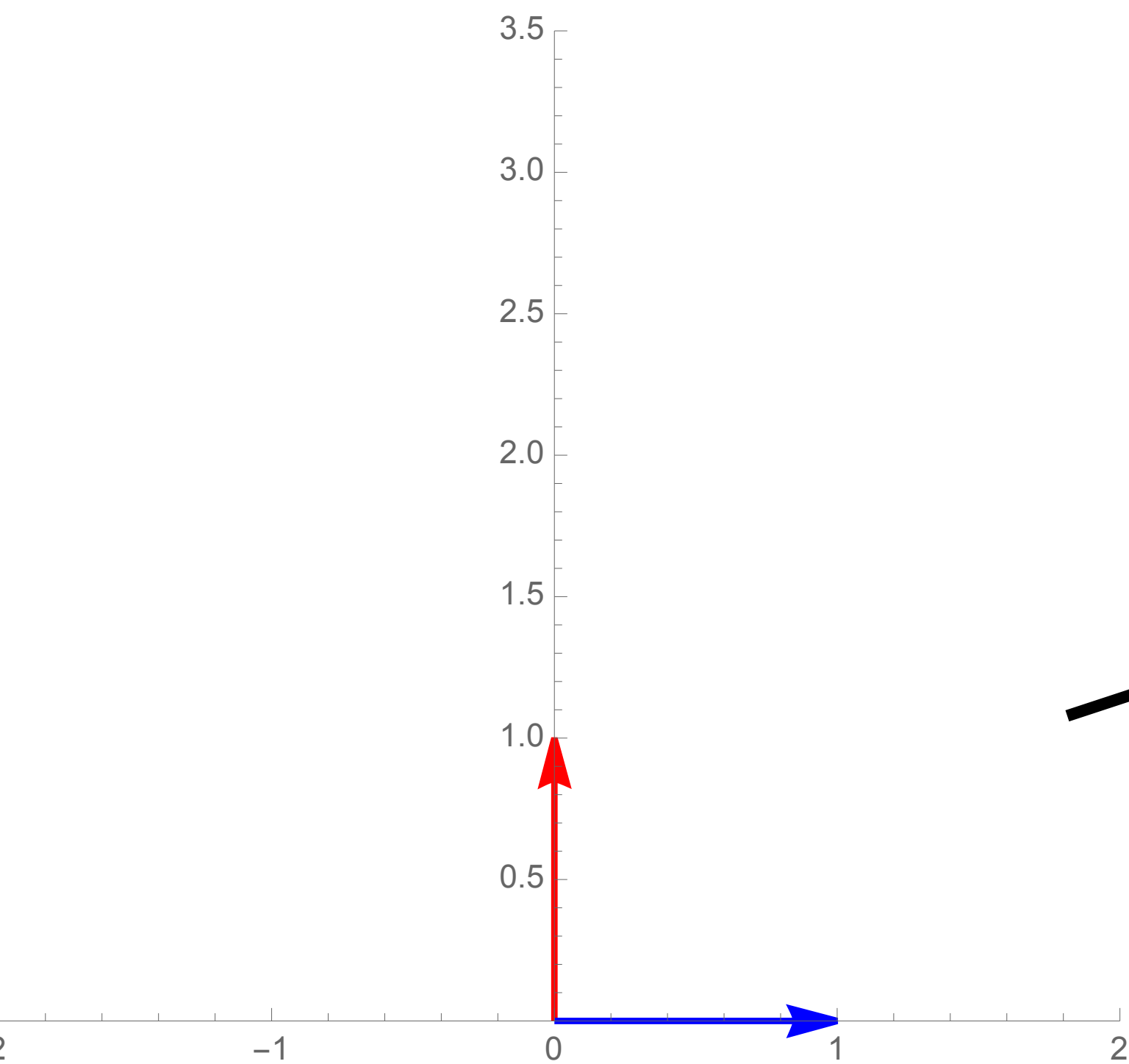
$$\begin{pmatrix} | & | & | \\ b_1 & \dots & b_n \\ | & | & | \end{pmatrix}$$

on the left.



$$\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

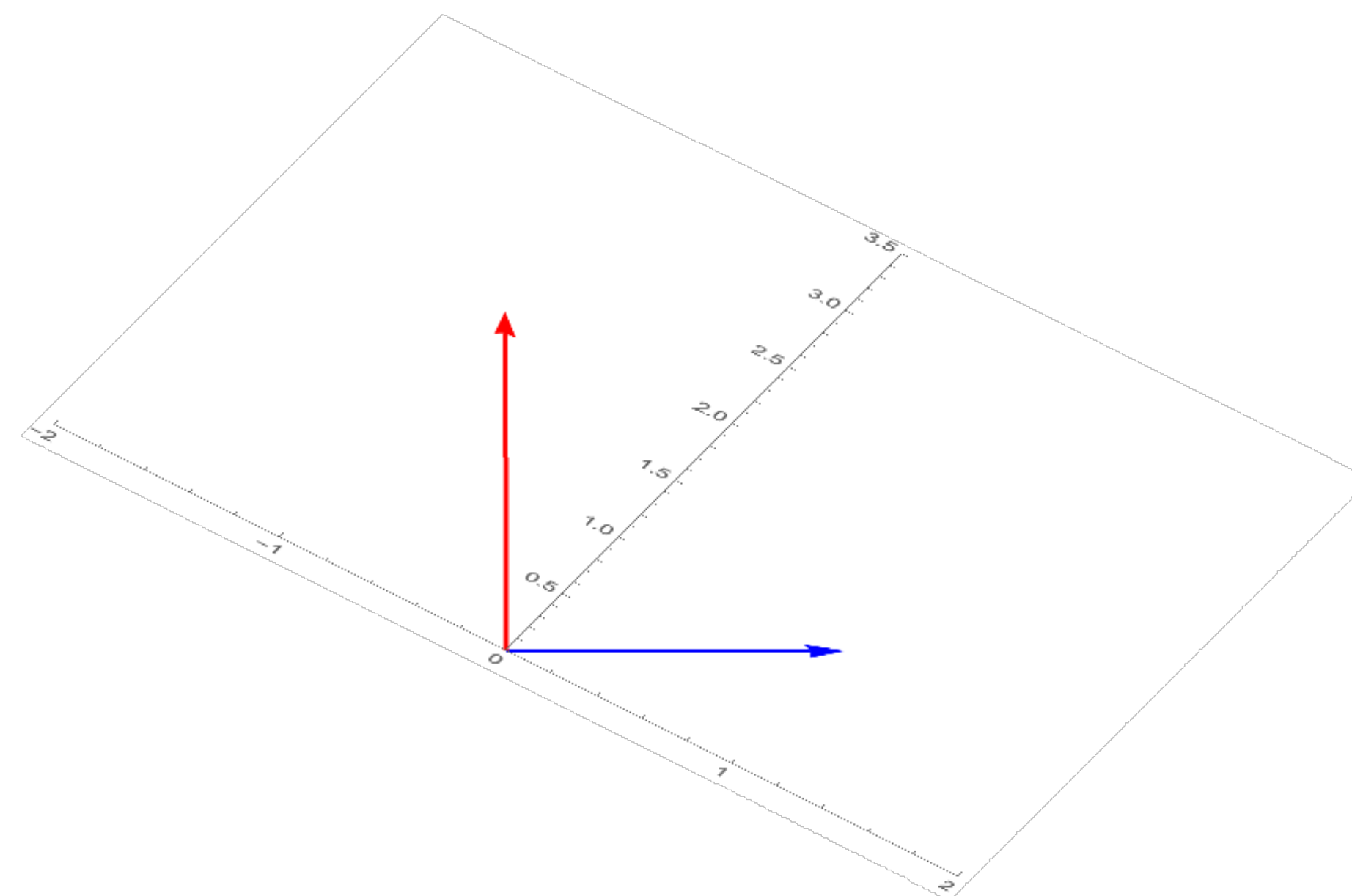
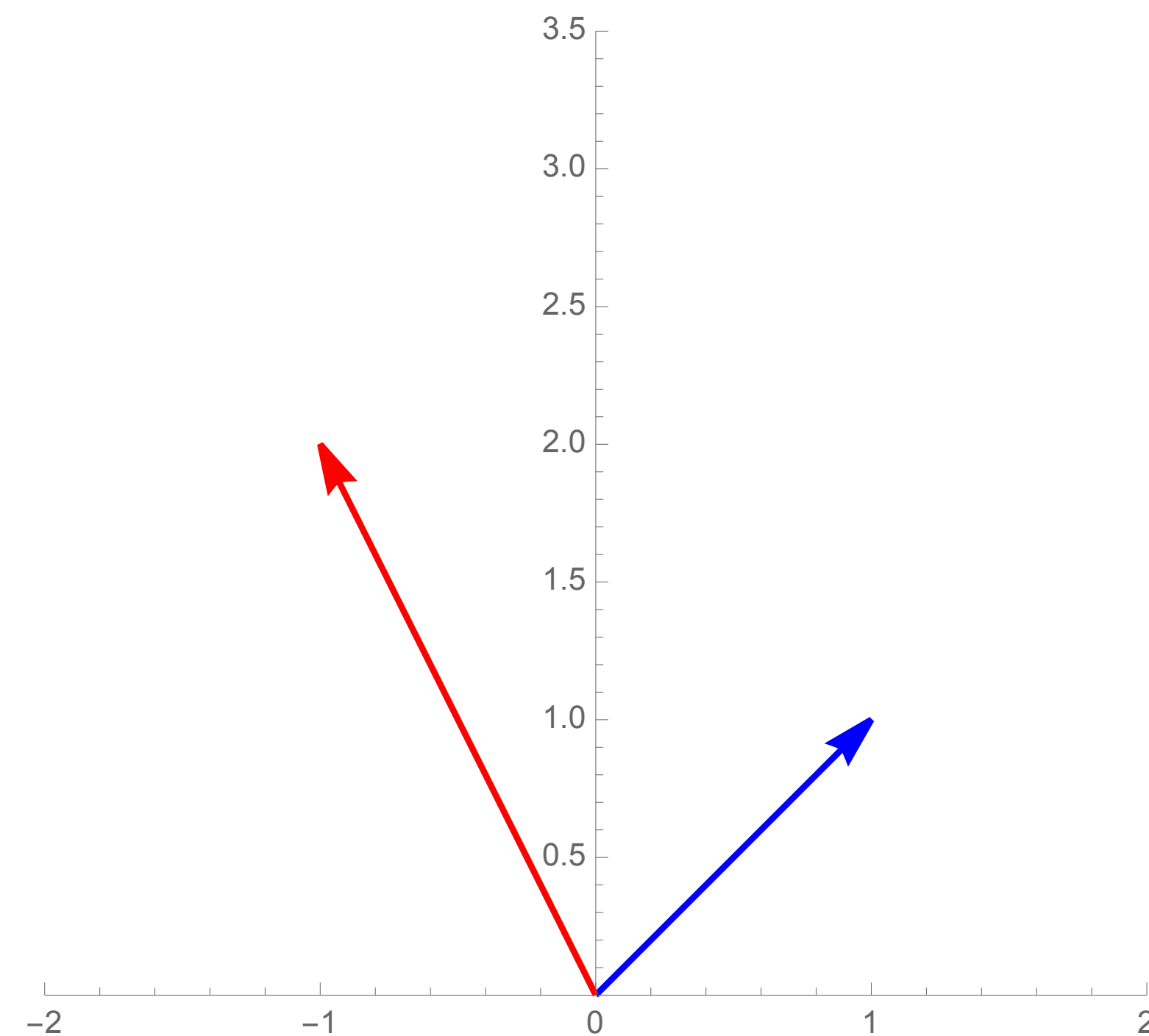




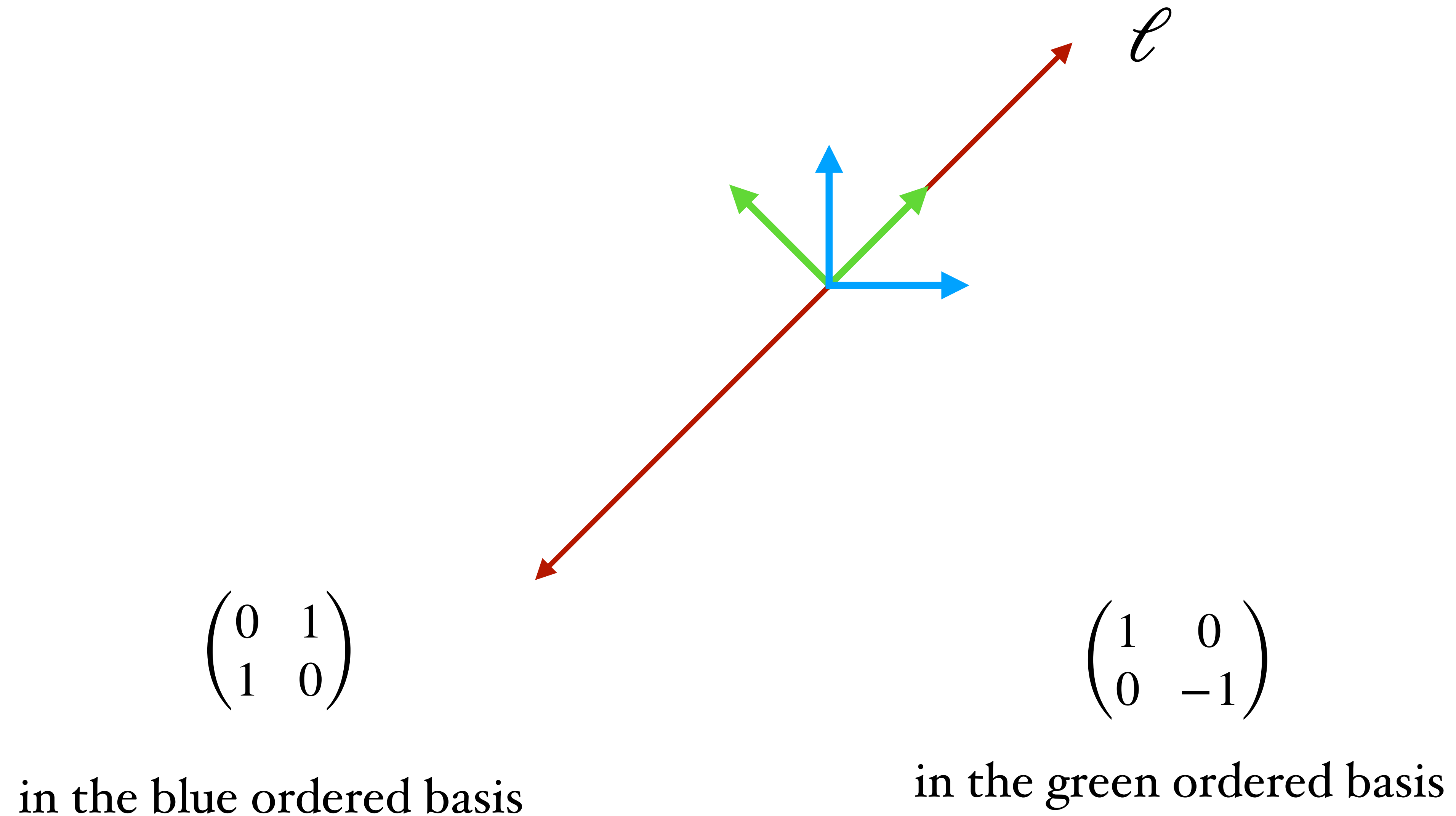
$$\begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}$$

alibi

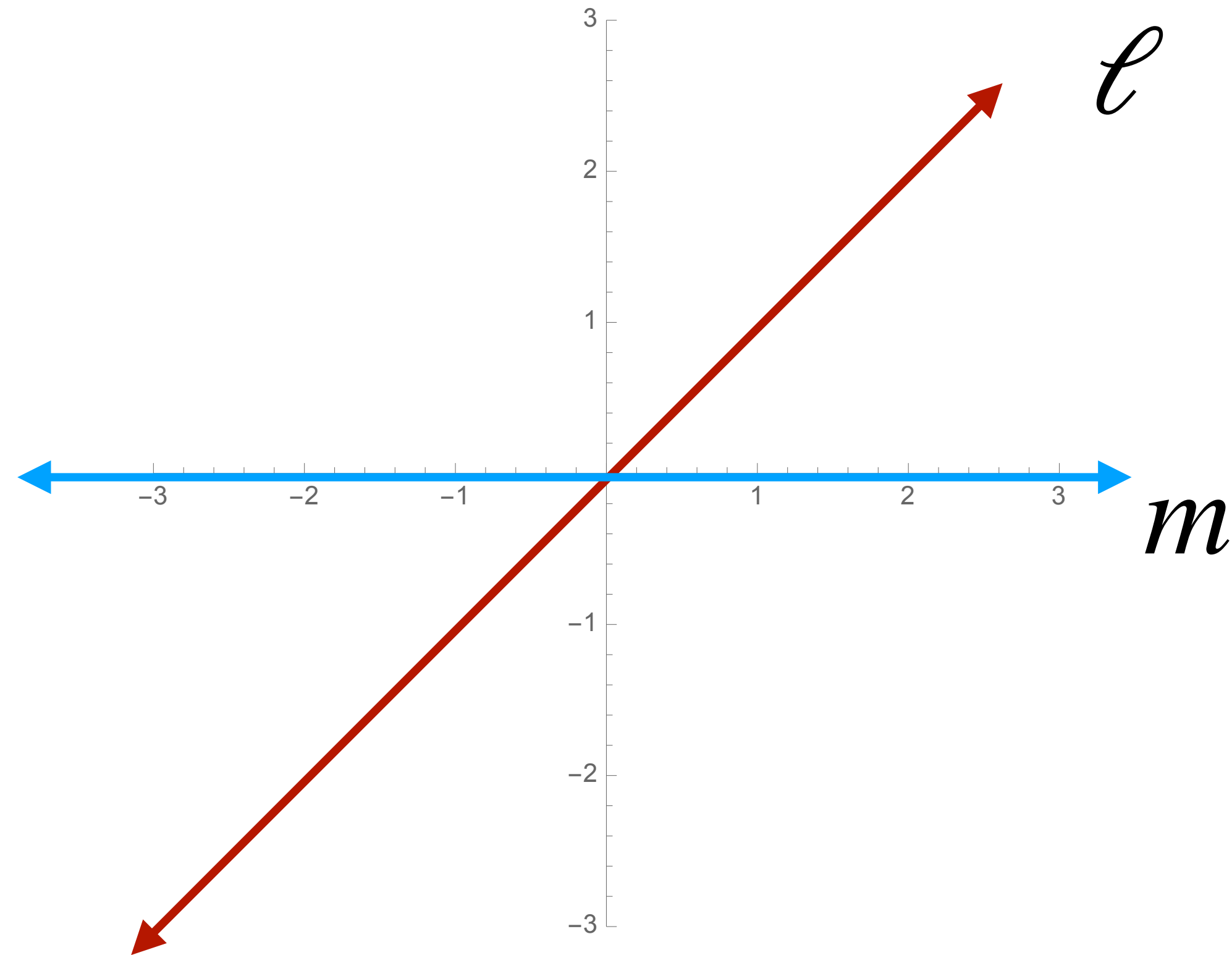
alias



Similar matrices represent the same linear transformation
in different coordinates.



Similar matrices represent different linear transformations that all have the same geometry but are located differently within the vector space, viewed with a fixed coordinate system.



$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

reflection in red line ℓ

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

reflection in blue line m

Similar matrices are related by a change of basis:

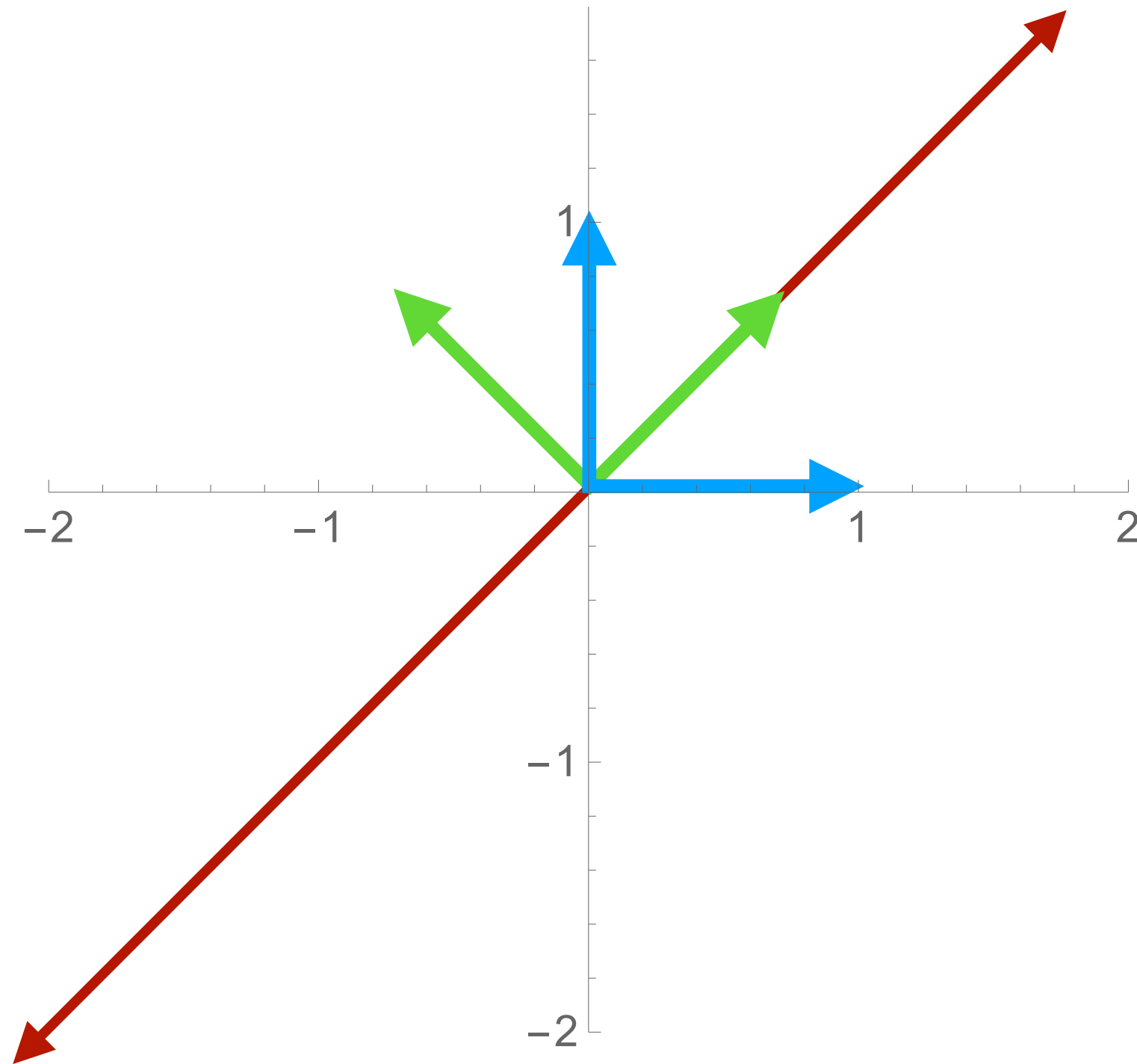
$$A \sim B \quad \text{if} \quad A = QBQ^{-1} \quad \text{for some invertible matrix } Q$$

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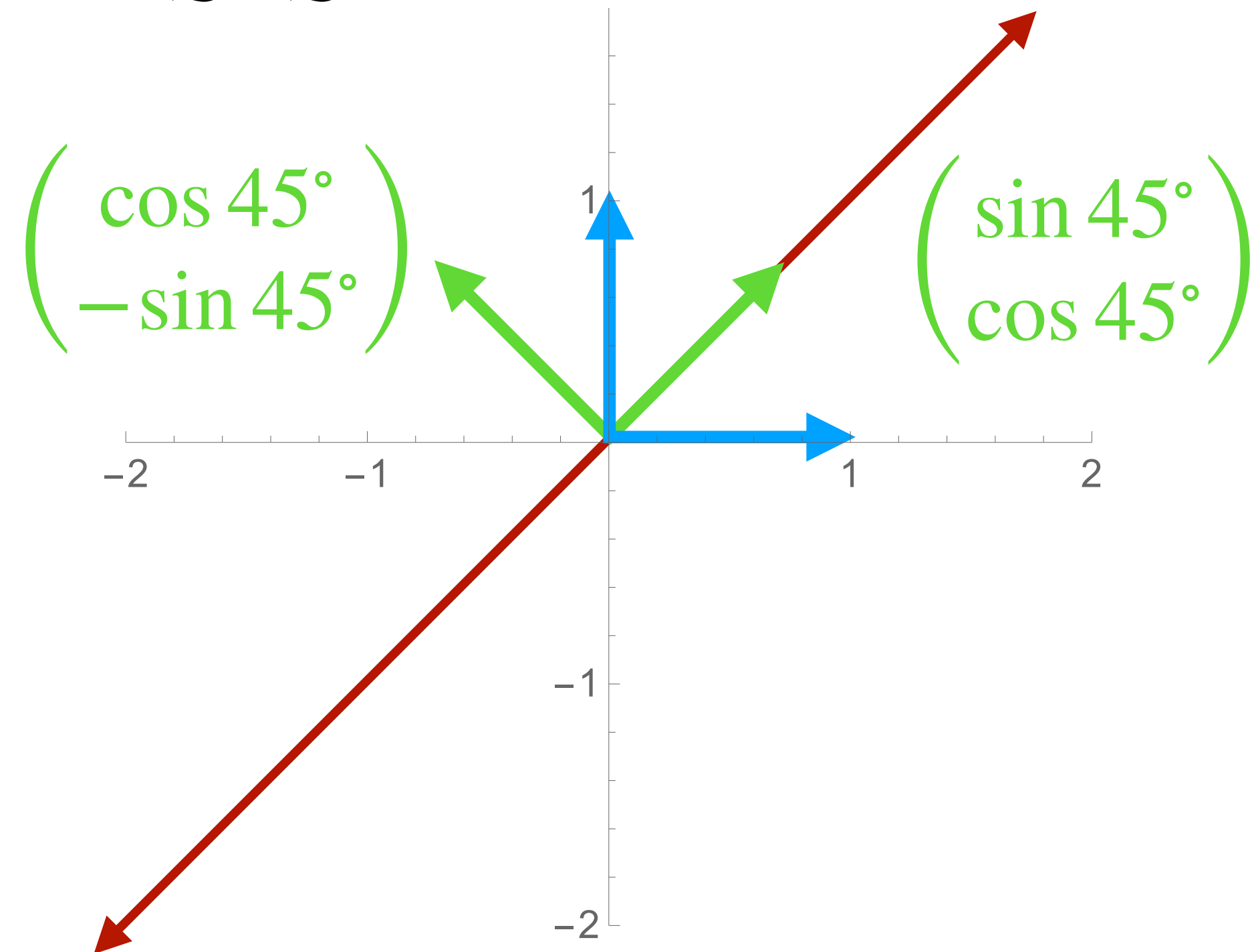


$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

in
blue basis
change back to
blue basis
in
green basis
change to
green basis

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in
blue basis

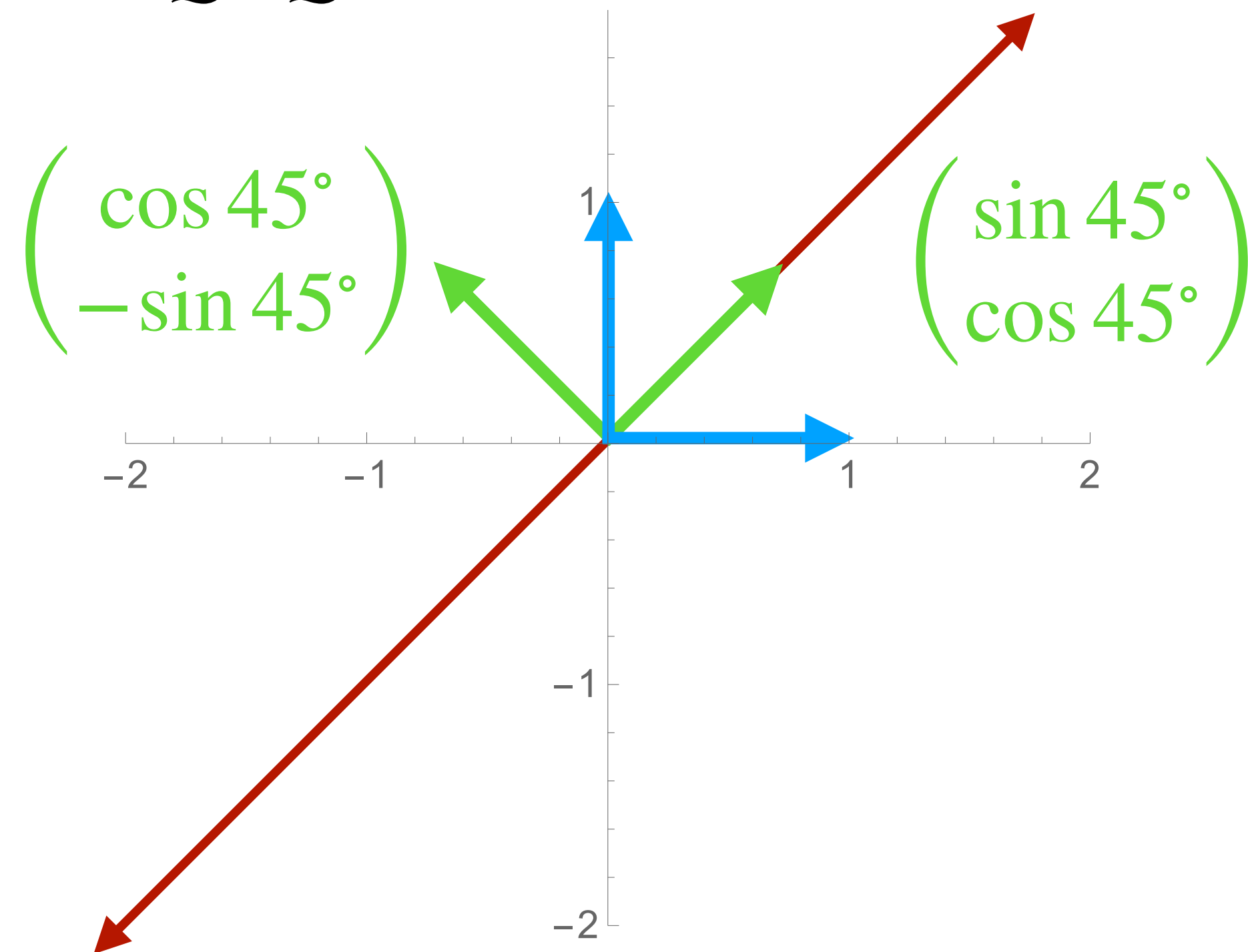
change back to
blue basis

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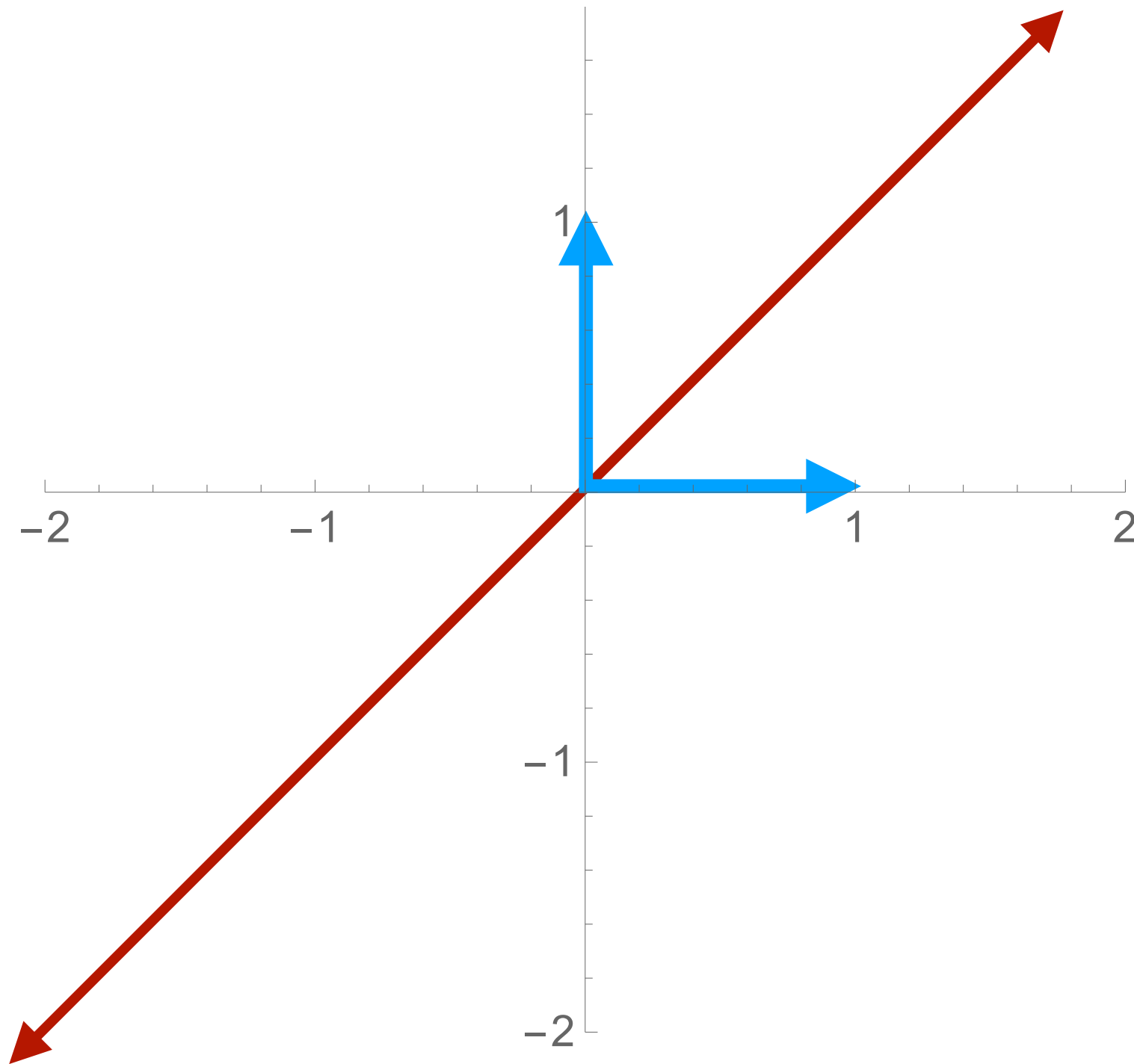


$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix}$$

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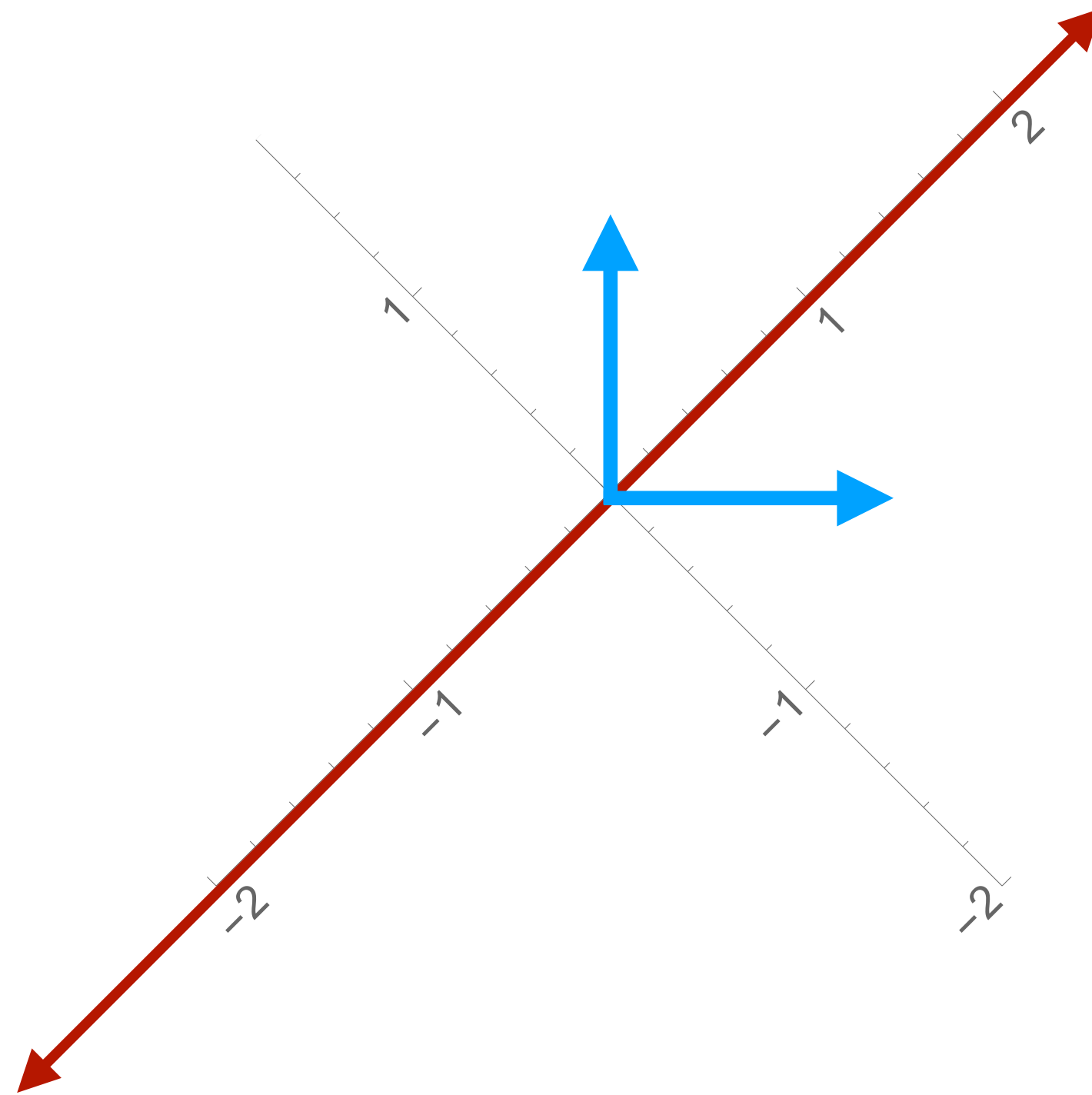


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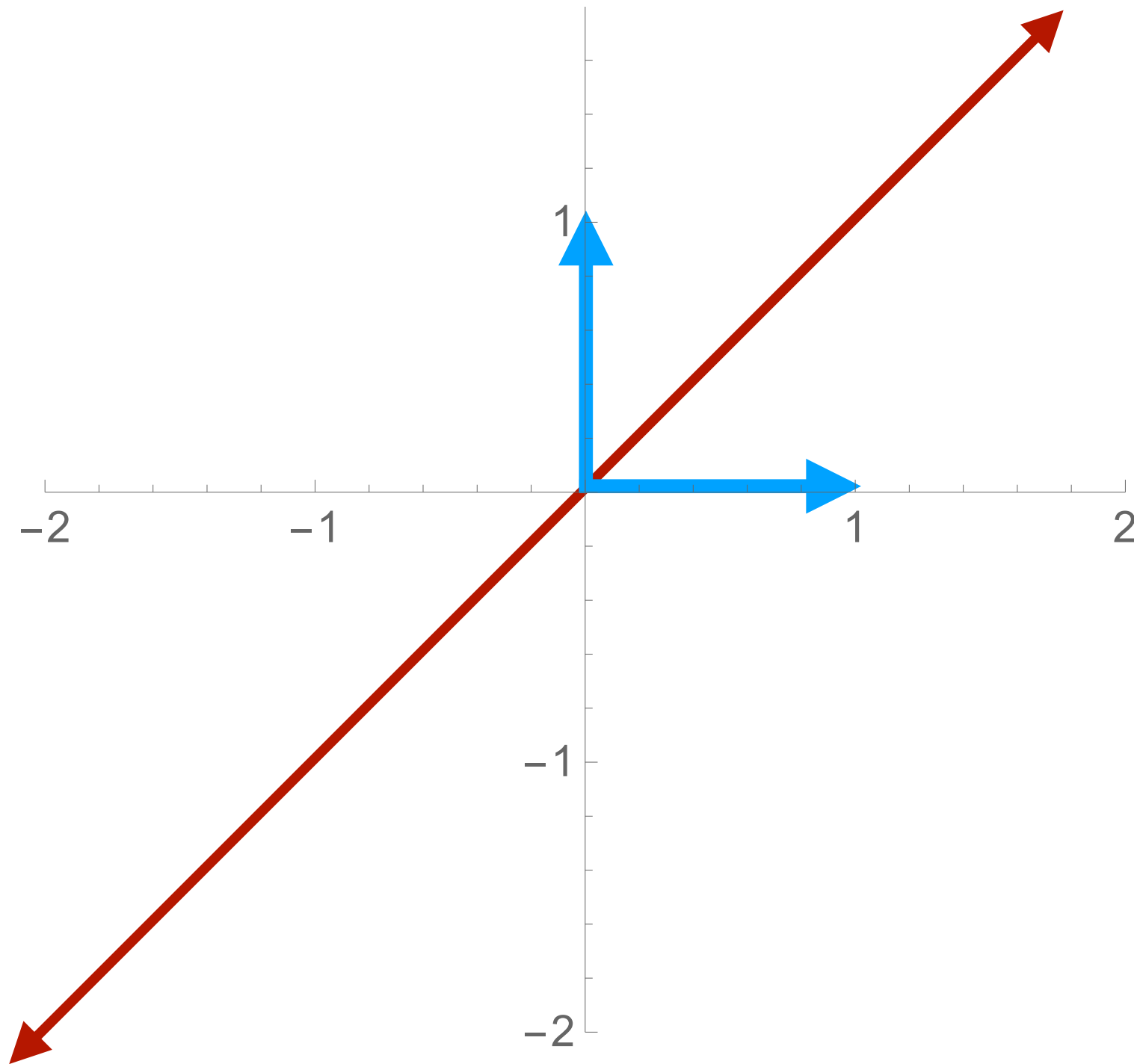


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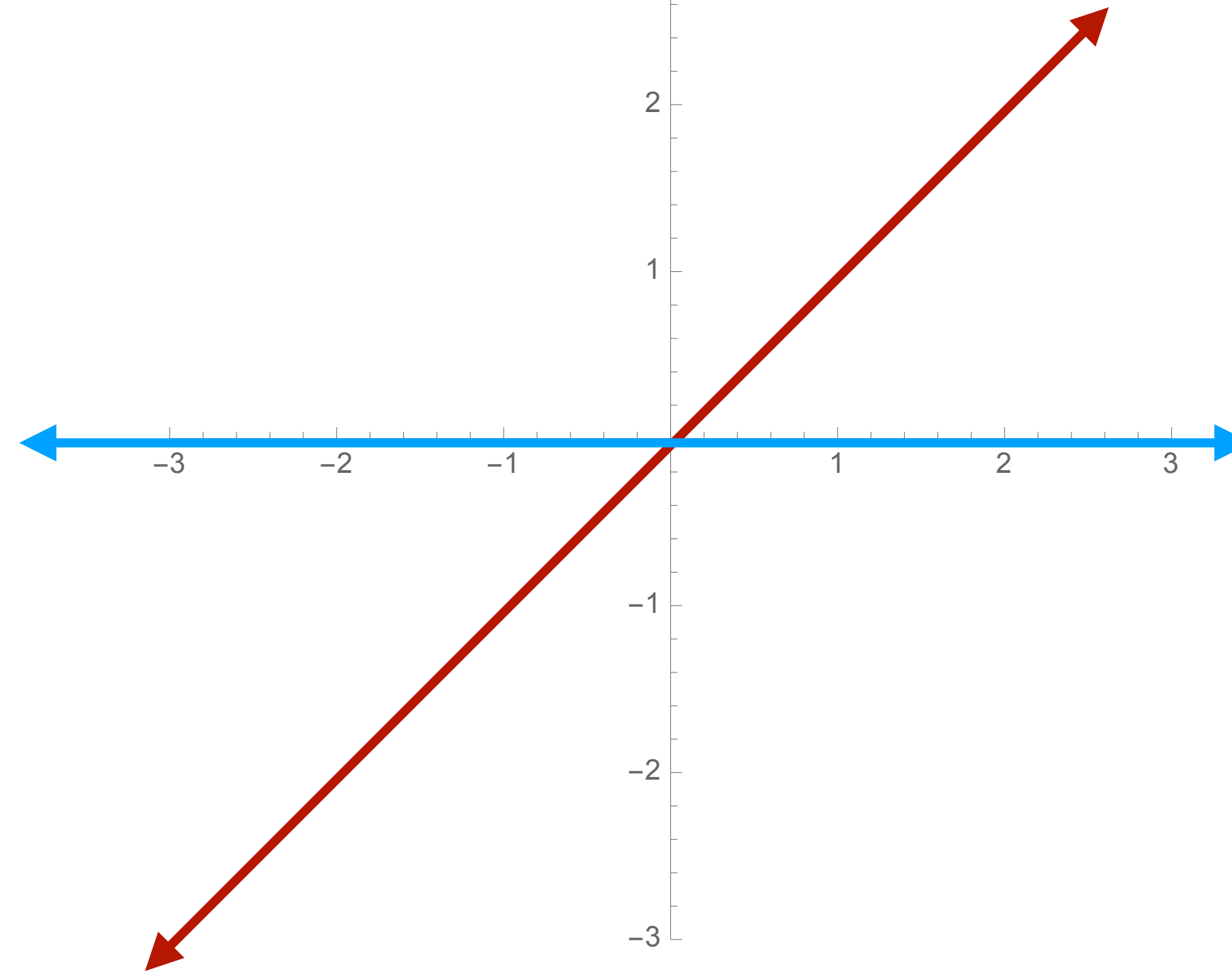


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Similar matrices are related by a geometric change of location:

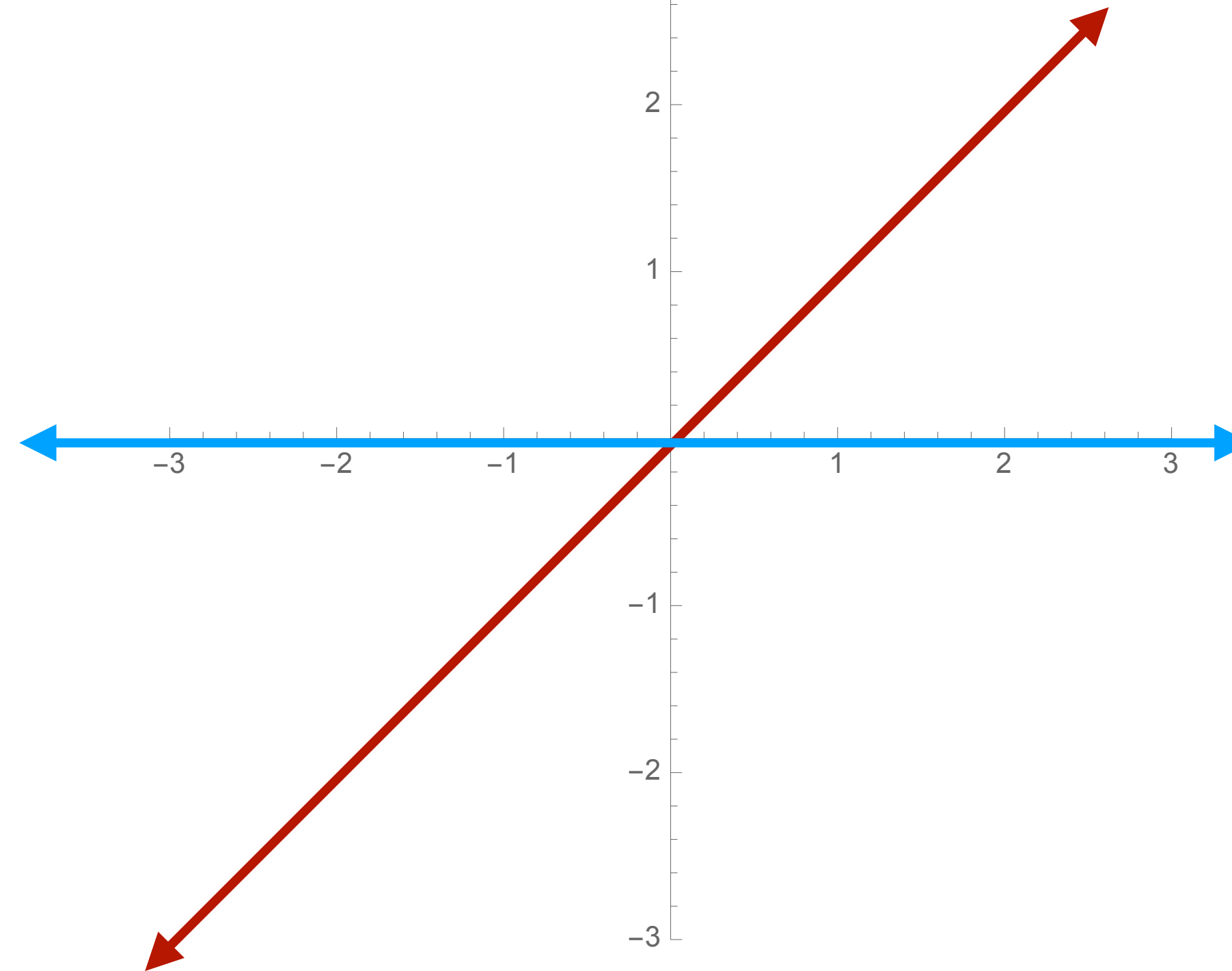
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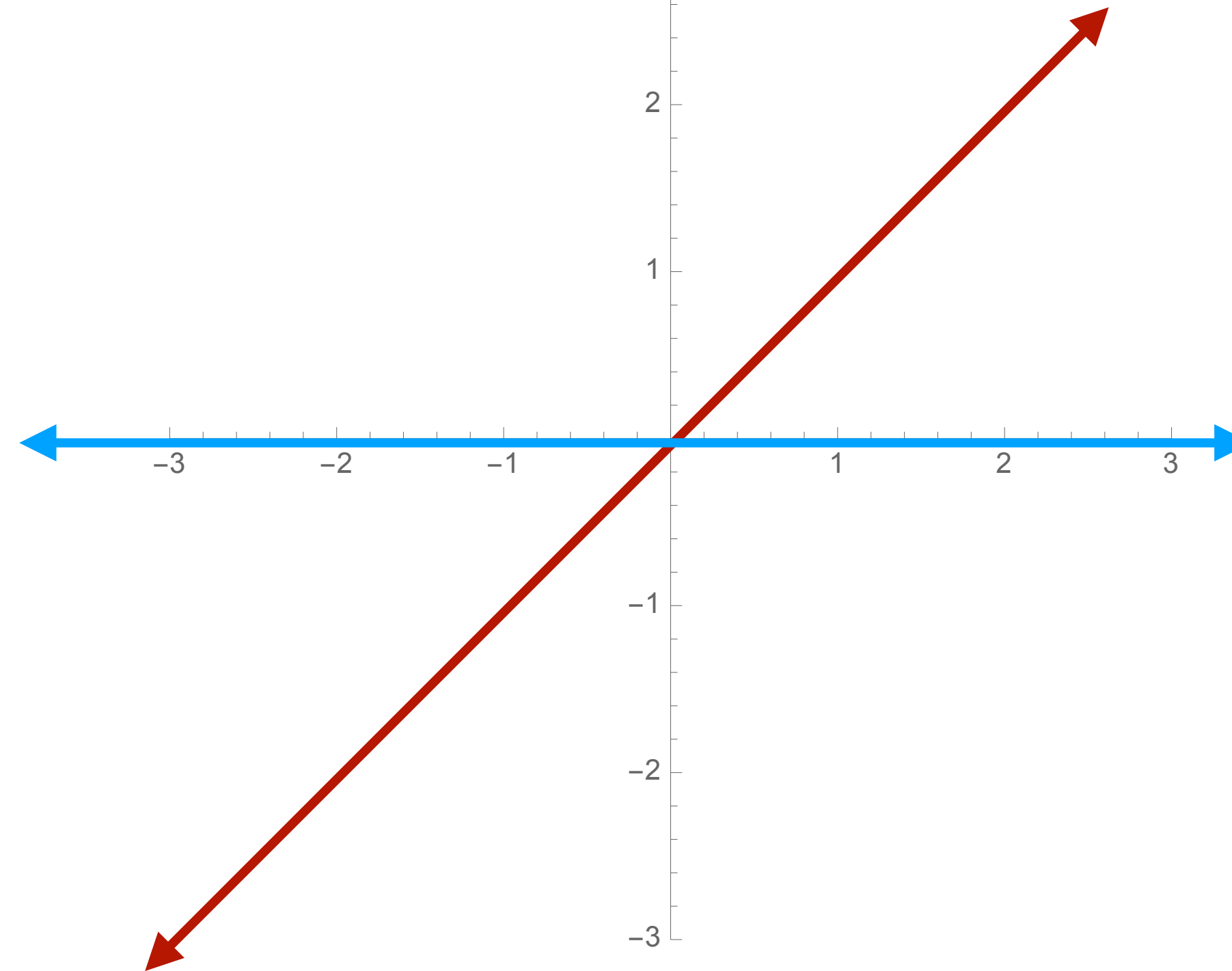
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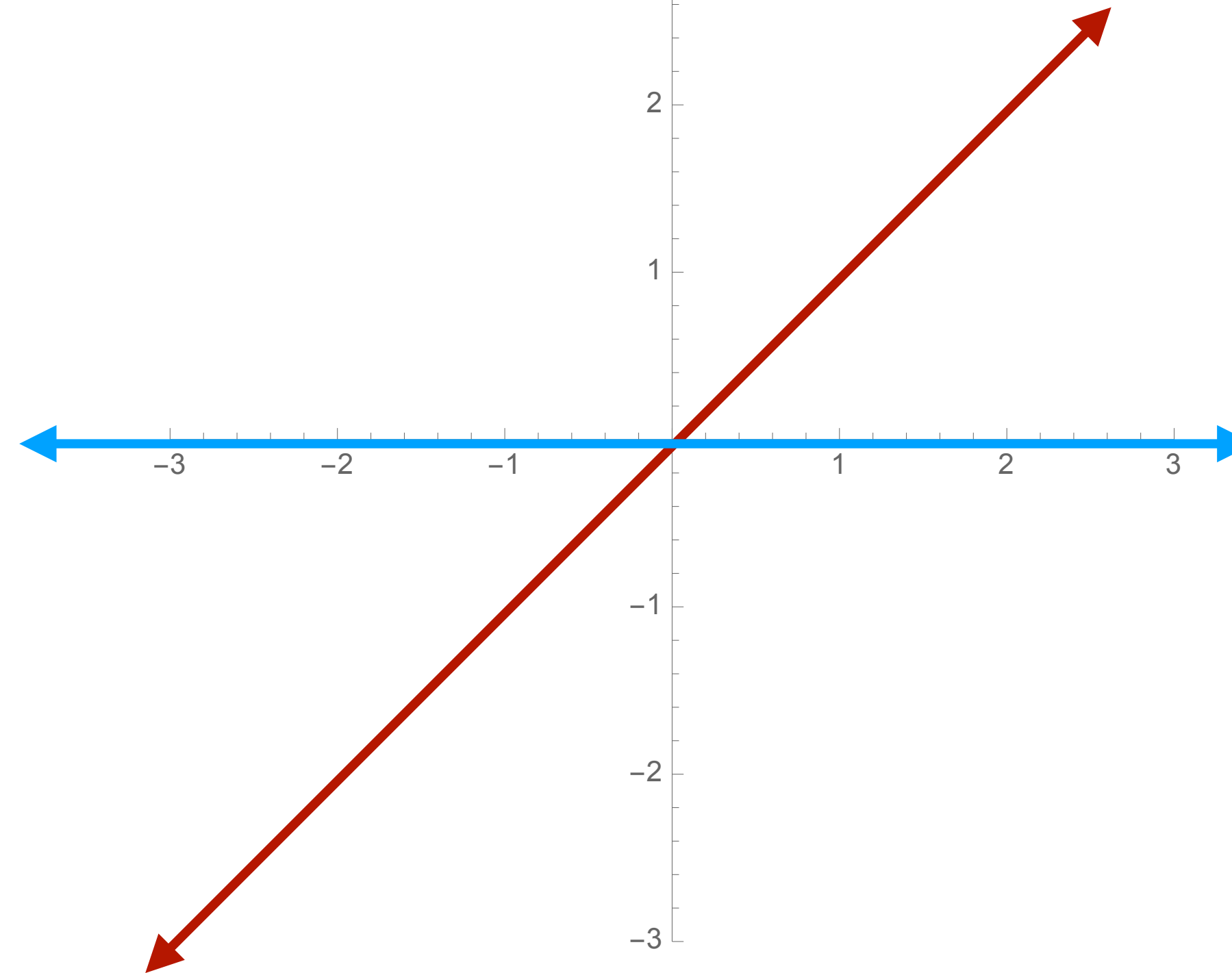
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reflection
in red line

reflection
in blue line

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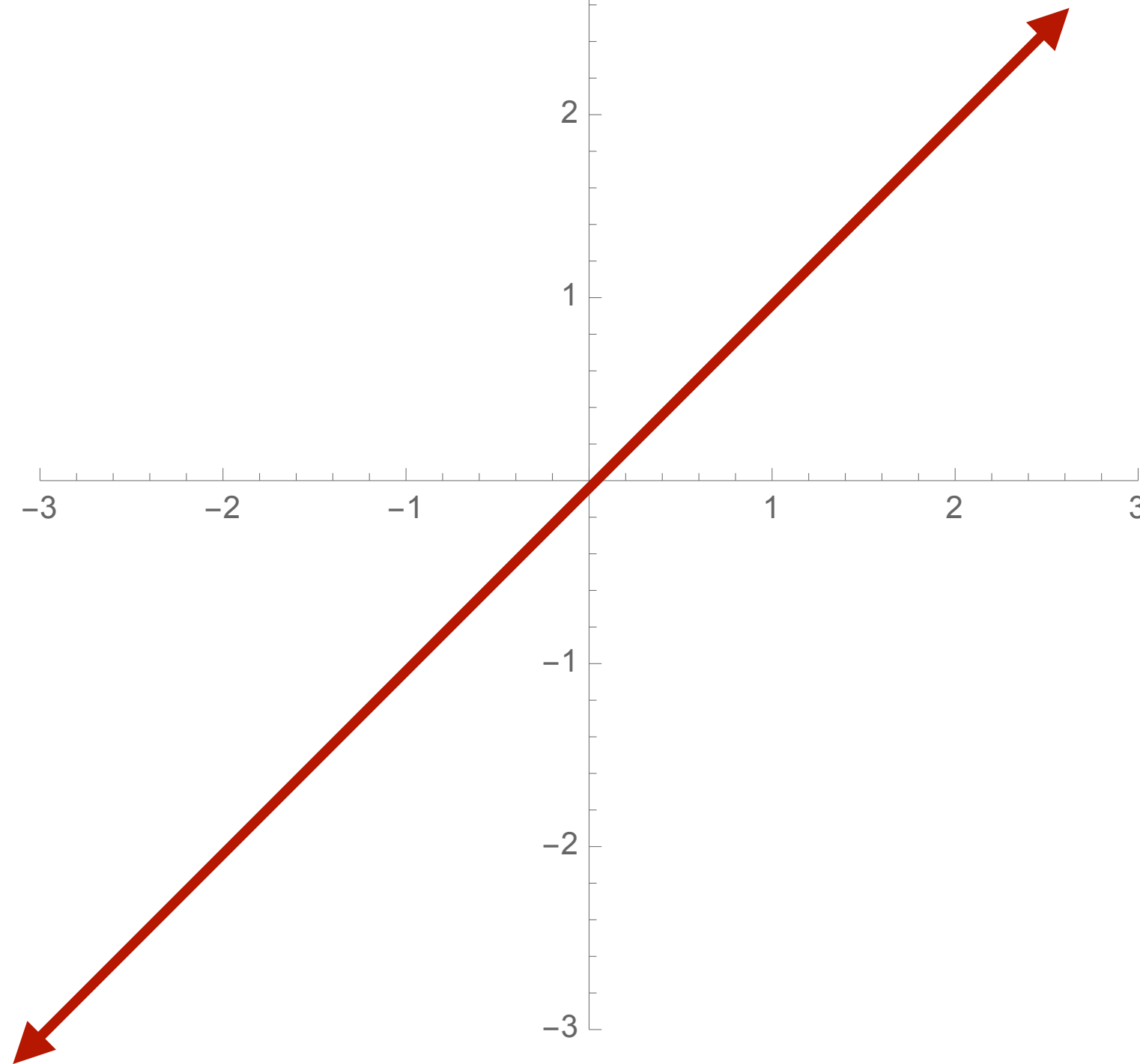


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reflection
rotate
reflection
rotate
in red line
blue to red
in blue line
red to blue

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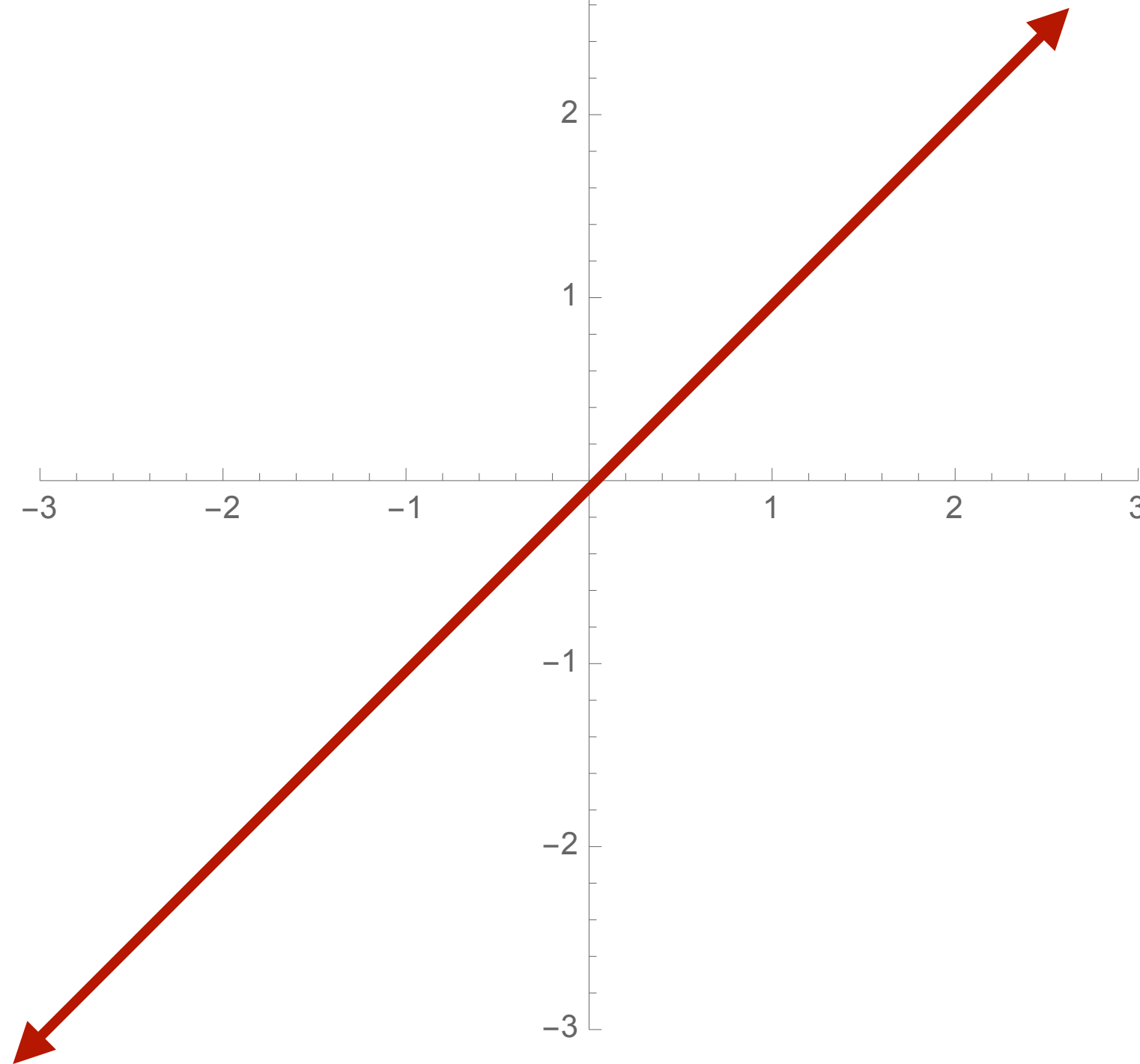
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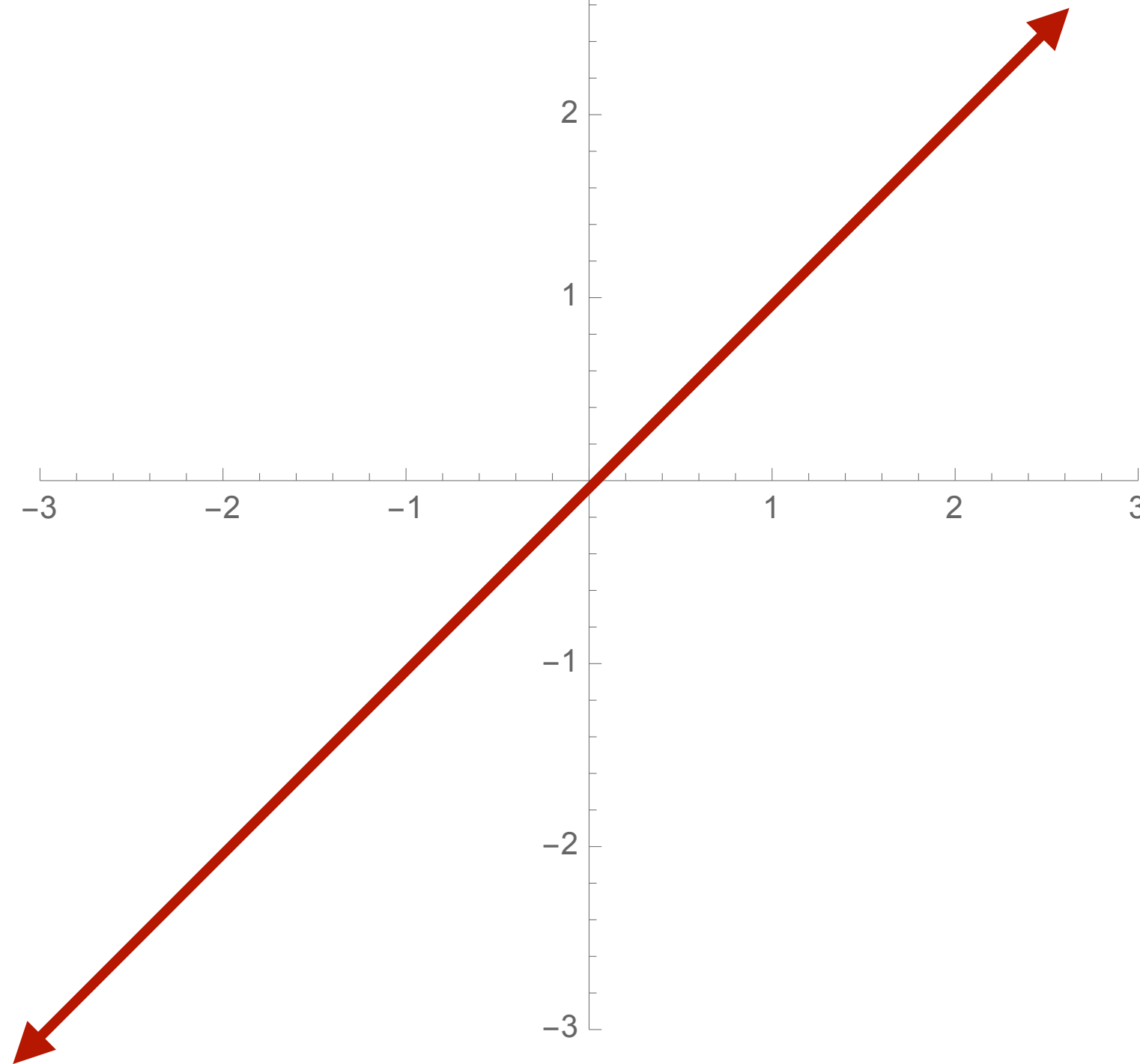
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rotates vectors
clockwise

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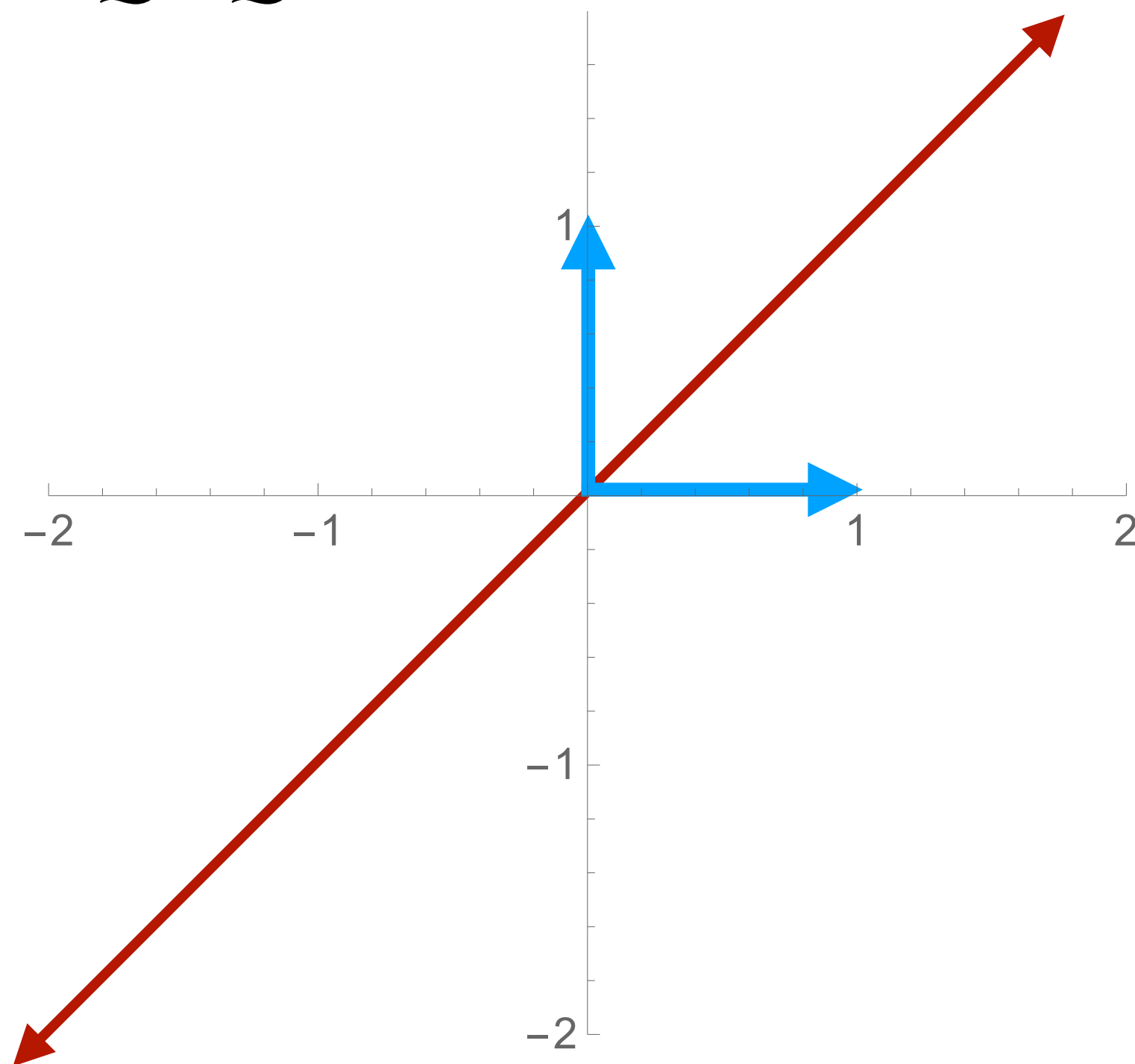
rotate
blue to red

reflection
in blue line

rotate
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Similar matrices are related by a change of basis:

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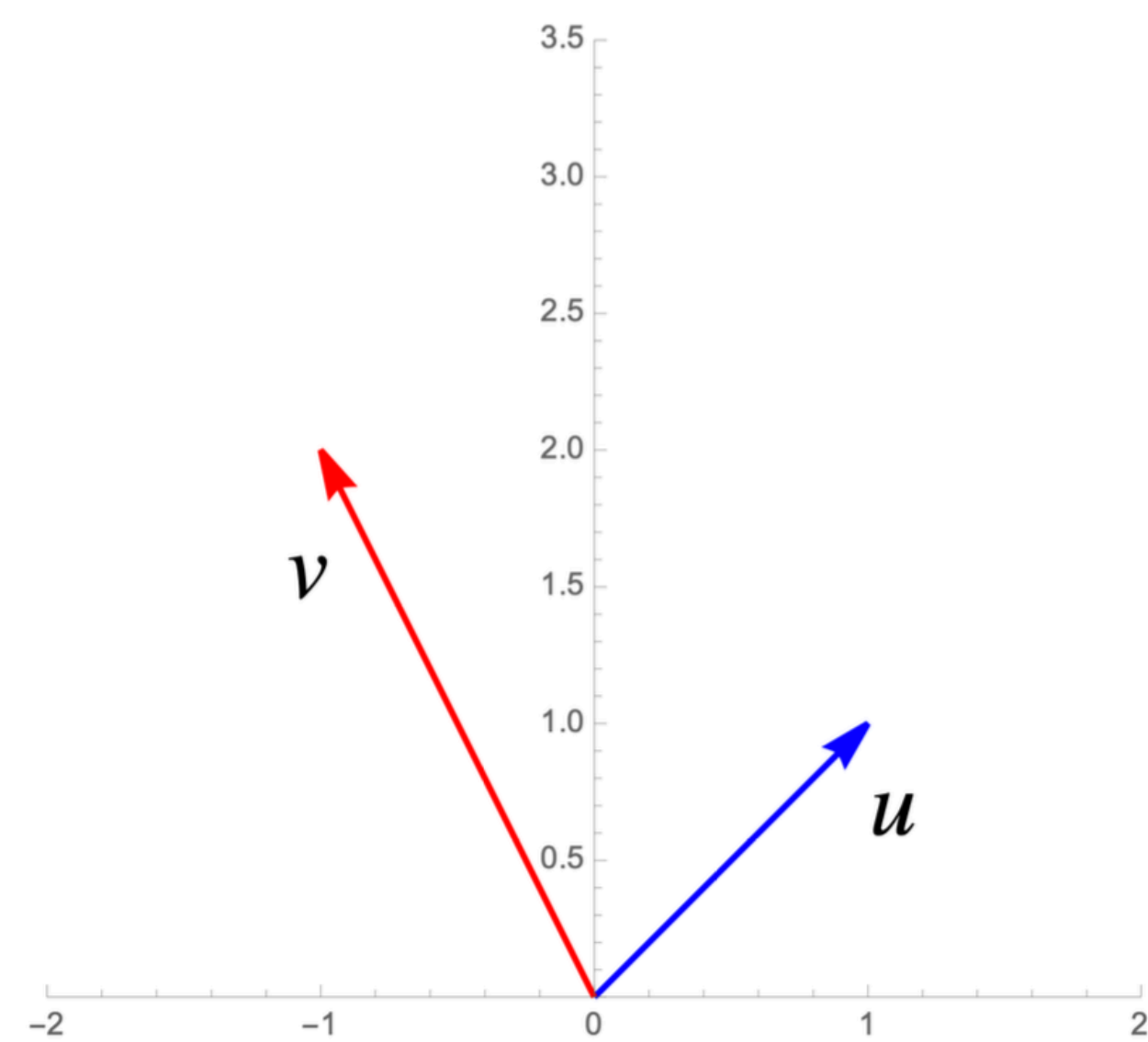
rotates coordinate system
counterclockwise

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} \cos 45^\circ & \sin 45^\circ \\ -\sin 45^\circ & \cos 45^\circ \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \cos 45^\circ & -\sin 45^\circ \\ \sin 45^\circ & \cos 45^\circ \end{pmatrix}$$

in
change back to
in
change to
blue basis
blue basis
green basis
green basis

During class 1

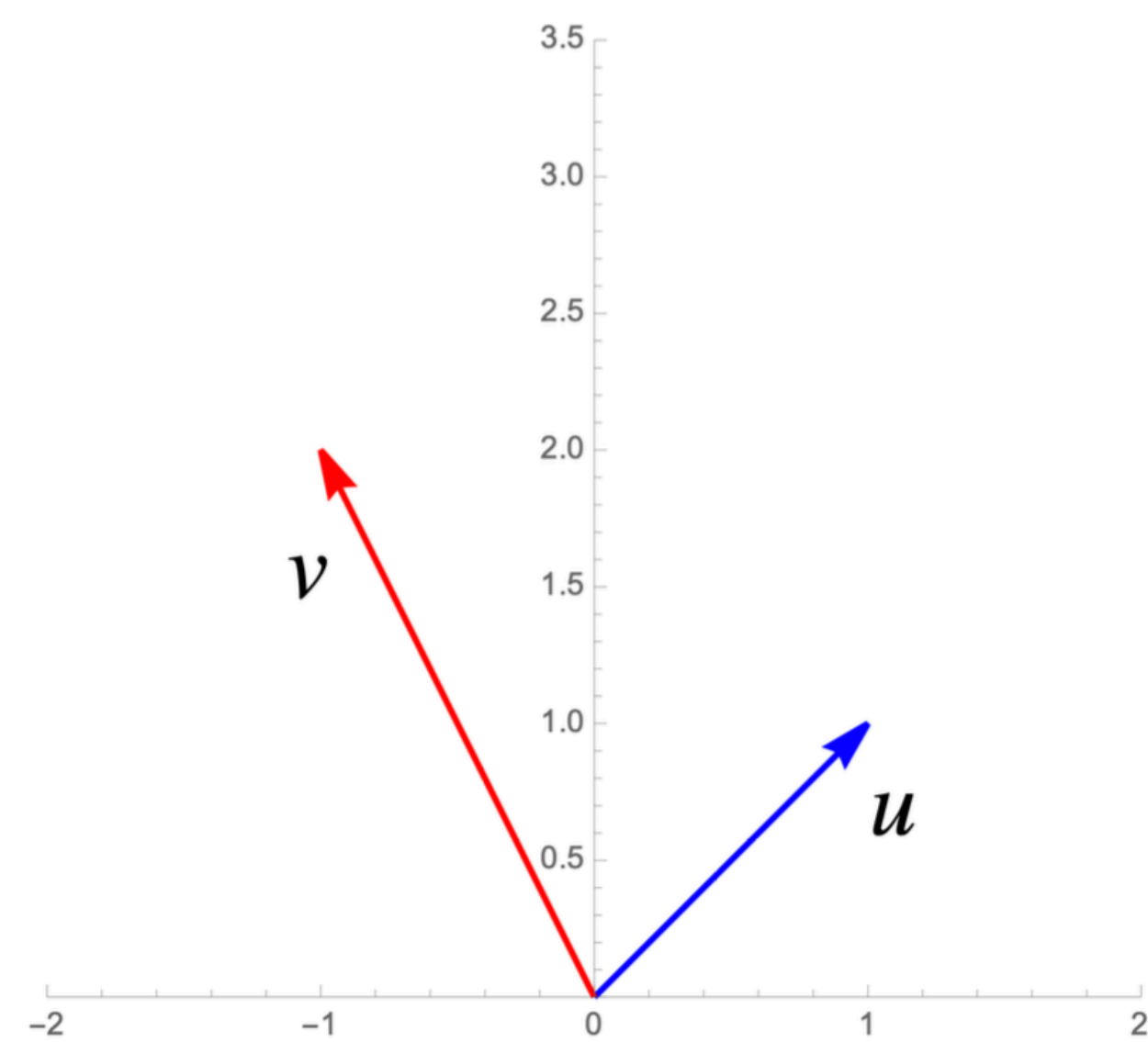
a) Using the ordered basis $u, v \in \mathbb{R}^2$ pictured below, write down the matrix representing the linear transformation that scales u by 2 and v by 3.



b) Now write down the matrix representing the same linear transformation, but using the standard basis for $\mathbb{R}^2$.

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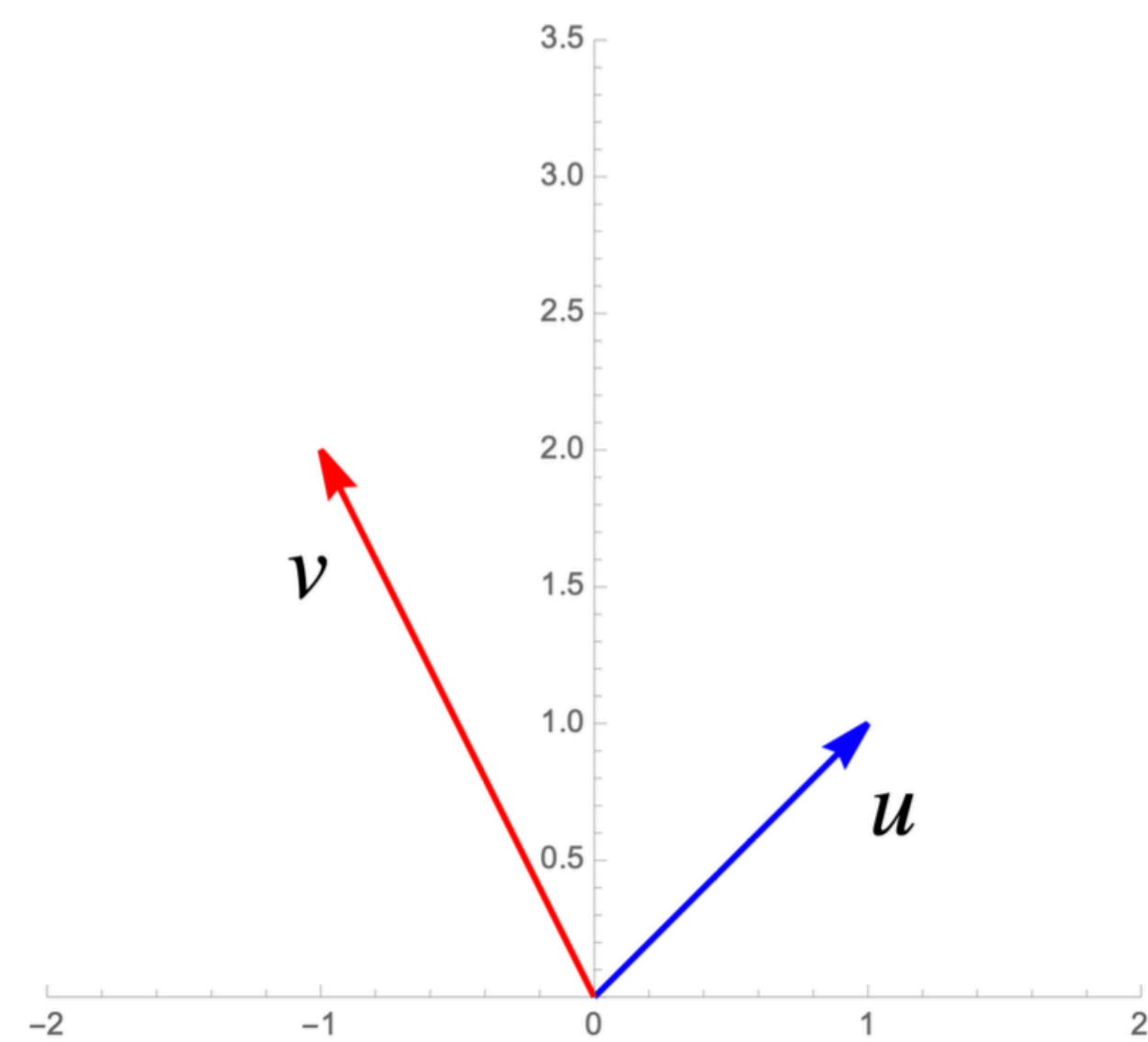


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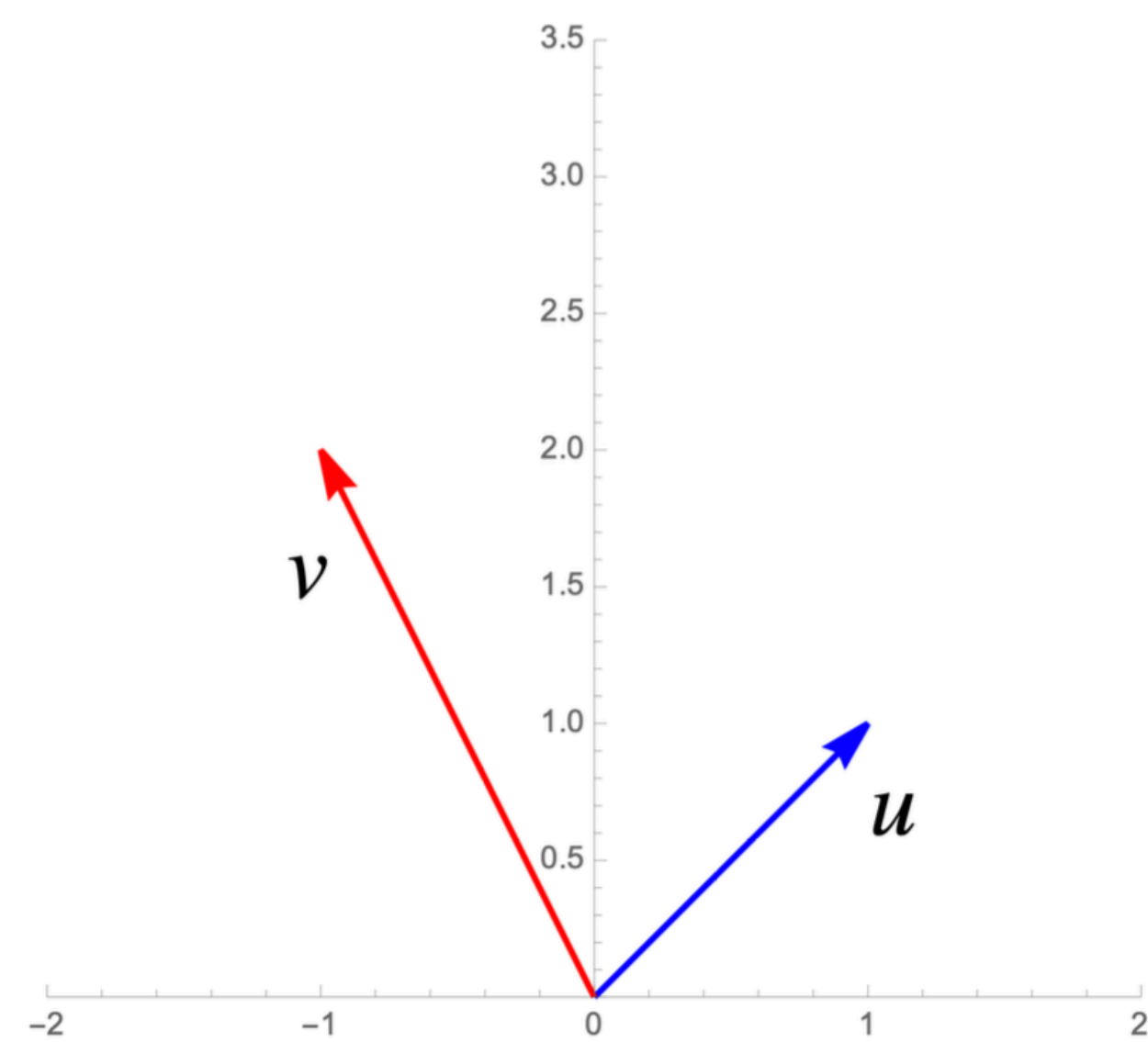
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$$? = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

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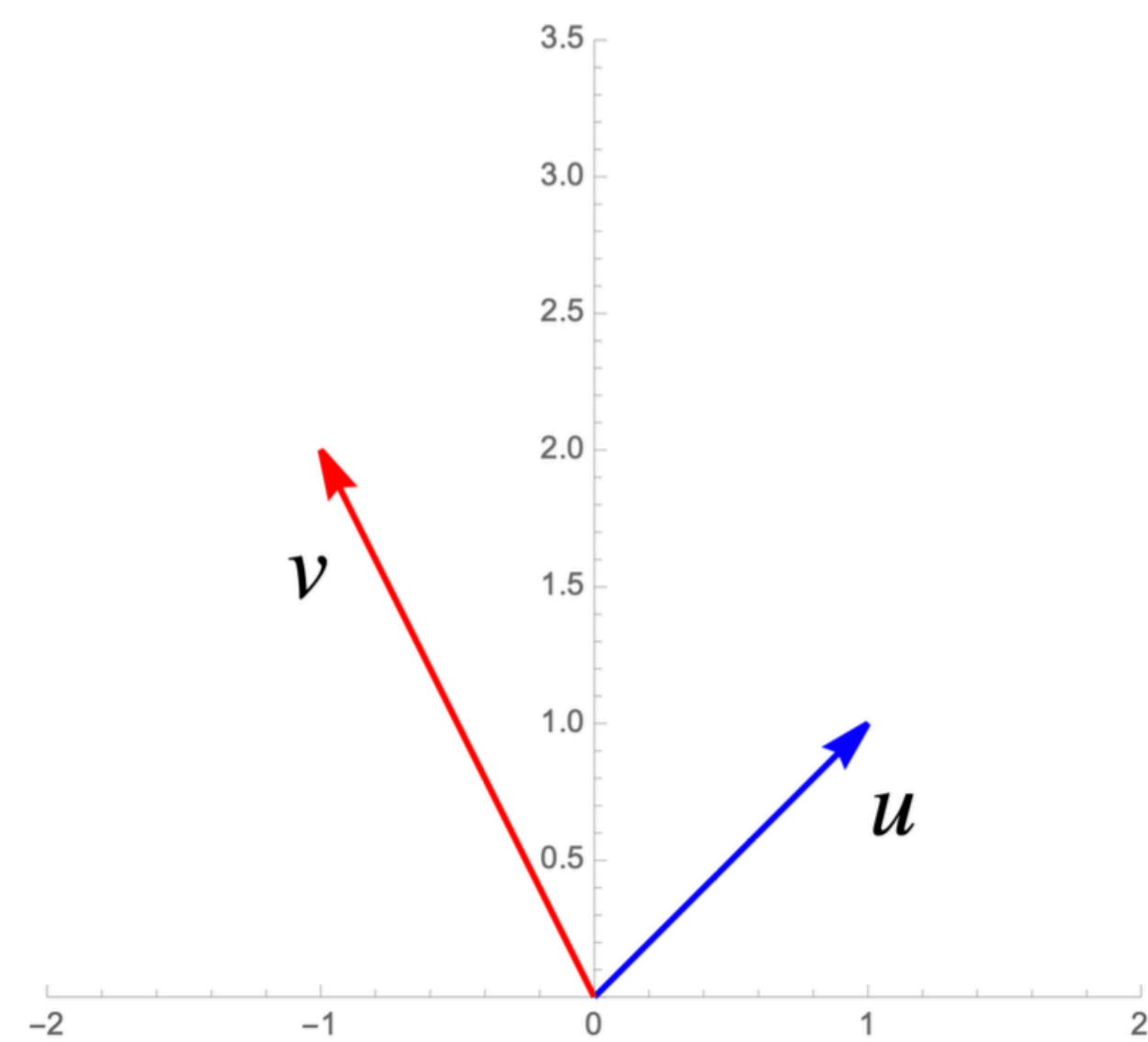
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a) Using the ordered basis $u, v \in \mathbb{R}^2$ pictured below, write down the matrix representing the linear transformation that scales u by 2 and v by 3.



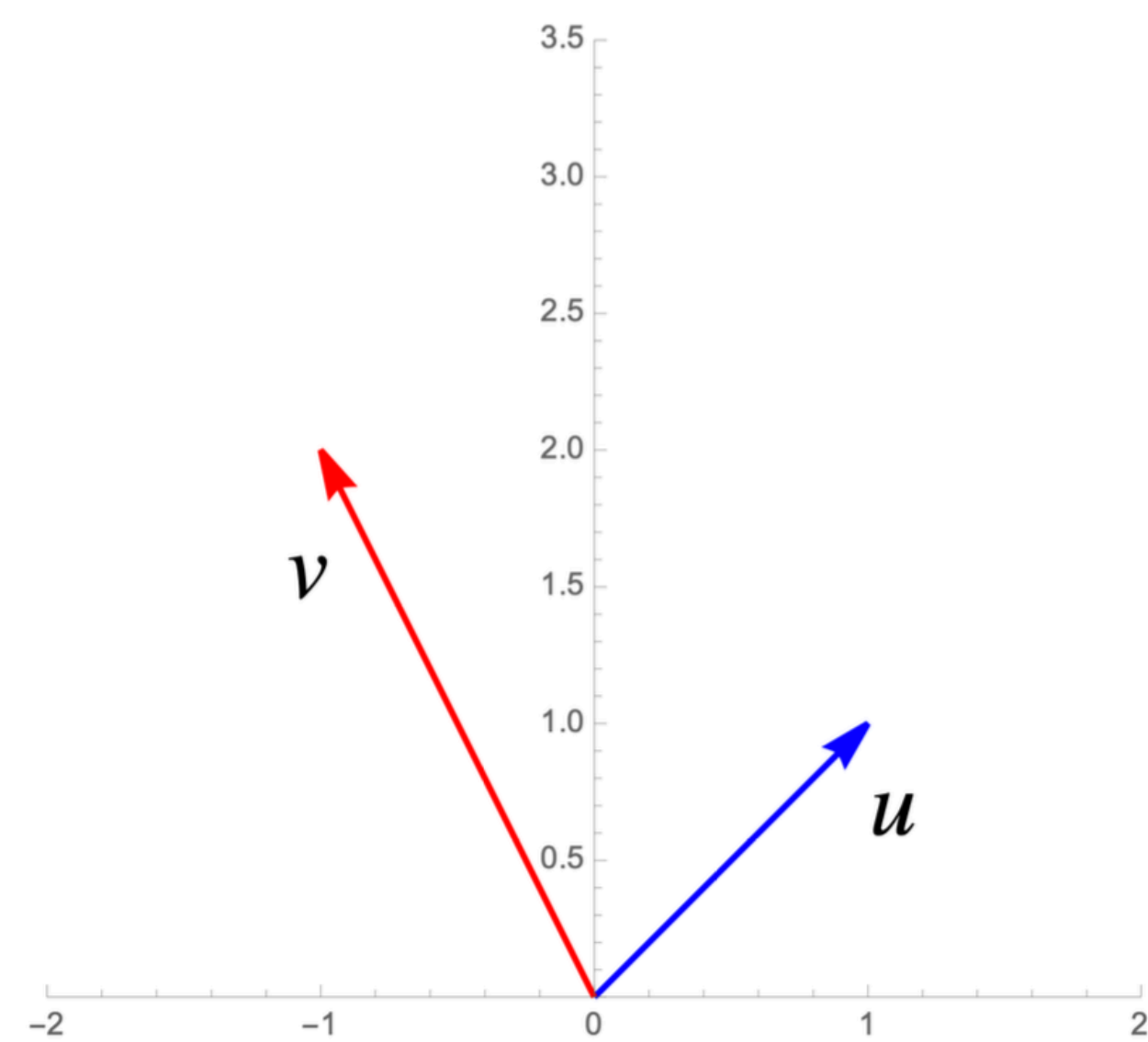
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b) Now write down the matrix representing the same linear transformation, but using the standard basis for $\mathbb{R}^2$.

$$\begin{pmatrix} 7/3 & -1/3 \\ -2/3 & 8/3 \end{pmatrix} = \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 1 & 2 \end{pmatrix}^{-1}$$

During class 1

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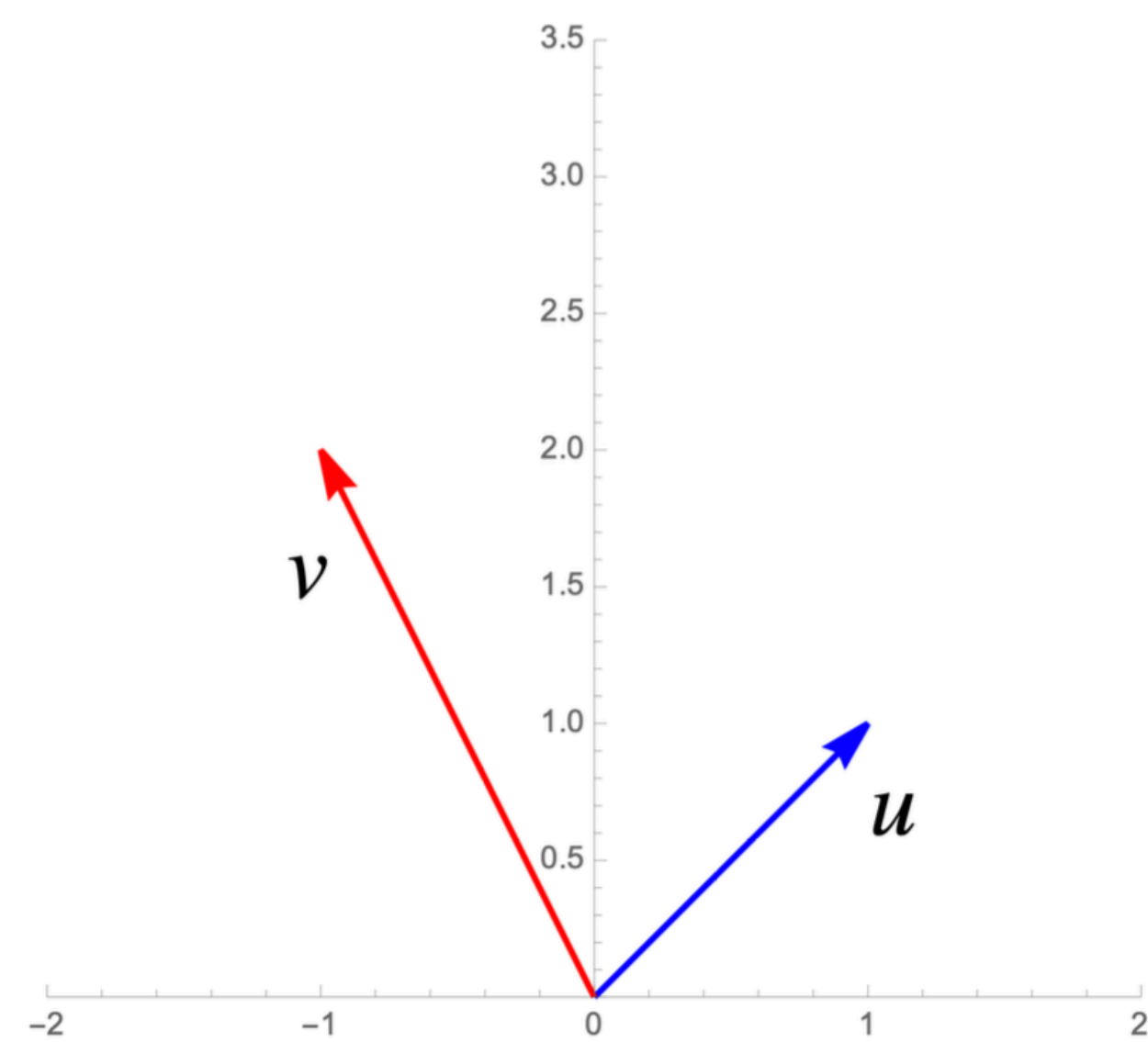
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$$\begin{pmatrix} 7/3 & -1/3 \\ -2/3 & 8/3 \end{pmatrix} \begin{pmatrix} -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 6 \end{pmatrix}$$

b) Now write down the matrix representing the same linear transformation, but using the standard basis for $\mathbb{R}^2$.

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There exists a unique map $M_{n \times n}^{\mathbb{F}} \rightarrow \mathbb{F}$ that is multilinear, antisymmetric and normalized, which is the determinant.

(Section 3.4)

Proof of existence and uniqueness, 2×2 case

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix}$$

Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & 0 \\ b & d \end{vmatrix} + \\ &\quad \begin{vmatrix} a & c \\ 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & 0 \end{vmatrix} \end{aligned}$$

Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= ad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + bd \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix} + \\ &\quad ac \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix} + bc \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \end{aligned}$$

Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= ad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \cancel{bd \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}} + \\ &\quad \cancel{ac \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}} + bc \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \end{aligned}$$

Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= ad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \cancel{bd \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}} + \\ &\quad \cancel{ac \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}} + bc \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \end{aligned}$$


Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= ad \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} + \cancel{bd \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}} + \\ &\quad \cancel{ac \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}} - bc \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \end{aligned}$$


Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= ad \cdot 1 + \cancel{bd \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}} + \\ &\quad \cancel{ac \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}} - bc \cdot 1 \end{aligned}$$

Proof of existence and uniqueness, 2×2 case

$$\begin{aligned} \begin{vmatrix} a & c \\ b & d \end{vmatrix} &= \begin{vmatrix} a & c \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & c \\ b & d \end{vmatrix} \\ &= ad \cdot 1 + \cancel{bd \begin{vmatrix} 0 & 0 \\ 1 & 1 \end{vmatrix}} + \\ &\quad \cancel{ac \begin{vmatrix} 1 & 1 \\ 0 & 0 \end{vmatrix}} - bc \cdot 1 = ad - bc \end{aligned}$$

Proof of existence and uniqueness, 3×3 case

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix}$$

Proof of existence and uniqueness, 3×3 case

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix}$$

Proof of existence and uniqueness, 3×3 case

$$\begin{aligned}
 \begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} &= \begin{vmatrix} a & d & g \\ 0 & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & e & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & e & h \\ c & f & i \end{vmatrix} \\
 &= \begin{vmatrix} a & d & g \\ 0 & 0 & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & 0 & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & 0 & h \\ c & 0 & i \end{vmatrix} + \\
 &\quad \begin{vmatrix} a & 0 & g \\ 0 & e & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ b & e & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ 0 & e & h \\ c & 0 & i \end{vmatrix} + \\
 &\quad \begin{vmatrix} a & 0 & g \\ 0 & 0 & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ b & 0 & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ 0 & 0 & h \\ c & f & i \end{vmatrix}
 \end{aligned}$$

Proof of existence and uniqueness, 3×3 case

$$\begin{aligned} &\begin{vmatrix} a & d & g \\ 0 & 0 & h \\ 0 & 0 & i \end{vmatrix} & + & \begin{vmatrix} 0 & d & g \\ b & 0 & h \\ 0 & 0 & i \end{vmatrix} & + & \begin{vmatrix} 0 & d & g \\ 0 & 0 & h \\ c & 0 & i \end{vmatrix} & + & \\ &\begin{vmatrix} a & 0 & g \\ 0 & e & h \\ 0 & 0 & i \end{vmatrix} & + & \begin{vmatrix} 0 & 0 & g \\ b & e & h \\ 0 & 0 & i \end{vmatrix} & + & \begin{vmatrix} 0 & 0 & g \\ 0 & e & h \\ c & 0 & i \end{vmatrix} & + & \\ &\begin{vmatrix} a & 0 & g \\ 0 & 0 & h \\ 0 & f & i \end{vmatrix} & + & \begin{vmatrix} 0 & 0 & g \\ b & 0 & h \\ 0 & f & i \end{vmatrix} & + & \begin{vmatrix} 0 & 0 & g \\ 0 & 0 & h \\ c & f & i \end{vmatrix} & + & \end{aligned}$$

Proof of existence and uniqueness, 3×3 case

$$\begin{aligned} &\begin{vmatrix} a & d & g \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & d & 0 \\ 0 & 0 & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & d & 0 \\ 0 & 0 & 0 \\ 0 & 0 & i \end{vmatrix} \begin{vmatrix} 0 & d & g \\ b & 0 & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & 0 & h \\ c & 0 & i \end{vmatrix} + \\ &\begin{vmatrix} a & 0 & g \\ 0 & e & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ b & e & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ 0 & e & h \\ c & 0 & i \end{vmatrix} + \\ &\begin{vmatrix} a & 0 & g \\ 0 & 0 & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ b & 0 & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ 0 & 0 & h \\ c & f & i \end{vmatrix} \end{aligned}$$

Proof of existence and uniqueness, 3×3 case

$$\begin{aligned} &\begin{vmatrix} a & d & g \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & d & 0 \\ 0 & 0 & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ b & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & 0 \\ b & 0 & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & g \\ 0 & 0 & h \\ c & 0 & i \end{vmatrix} + \\ &\begin{vmatrix} a & 0 & g \\ 0 & e & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ b & e & h \\ 0 & 0 & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ 0 & e & h \\ c & 0 & i \end{vmatrix} + \\ &\begin{vmatrix} a & 0 & g \\ 0 & 0 & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ b & 0 & h \\ 0 & f & i \end{vmatrix} + \begin{vmatrix} 0 & 0 & g \\ 0 & 0 & h \\ c & f & i \end{vmatrix} \end{aligned}$$

Proof of existence and uniqueness, 3×3 case

$\begin{vmatrix} a & d & g \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & d & 0 \\ 0 & 0 & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & d & 0 \\ 0 & 0 & 0 \\ 0 & 0 & i \end{vmatrix}$	$\begin{vmatrix} 0 & d & g \\ b & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & 0 \\ b & 0 & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & 0 \\ b & 0 & 0 \\ 0 & 0 & i \end{vmatrix}$	$\begin{vmatrix} 0 & d & g \\ 0 & 0 & 0 \\ c & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & 0 \\ 0 & 0 & h \\ c & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & d & 0 \\ 0 & 0 & 0 \\ c & 0 & i \end{vmatrix}$
$\begin{vmatrix} a & 0 & g \\ 0 & e & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & 0 & 0 \\ 0 & e & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} a & 0 & 0 \\ 0 & e & 0 \\ 0 & 0 & i \end{vmatrix}$	$\begin{vmatrix} 0 & 0 & g \\ b & e & 0 \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ b & e & h \\ 0 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ b & e & 0 \\ 0 & 0 & i \end{vmatrix}$	$\begin{vmatrix} 0 & 0 & g \\ 0 & e & 0 \\ c & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ 0 & e & h \\ c & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ 0 & e & 0 \\ c & 0 & i \end{vmatrix}$
$\begin{vmatrix} a & 0 & g \\ 0 & 0 & 0 \\ 0 & f & 0 \end{vmatrix} + \begin{vmatrix} a & 0 & 0 \\ 0 & 0 & h \\ 0 & f & 0 \end{vmatrix} + \begin{vmatrix} a & 0 & 0 \\ 0 & 0 & 0 \\ 0 & f & i \end{vmatrix}$	$\begin{vmatrix} 0 & 0 & g \\ b & 0 & 0 \\ 0 & f & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ b & 0 & h \\ 0 & f & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ b & 0 & 0 \\ 0 & f & i \end{vmatrix}$	$\begin{vmatrix} 0 & 0 & g \\ 0 & 0 & 0 \\ c & f & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & h \\ c & f & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ c & f & i \end{vmatrix}$

Proof of existence and uniqueness, 3×3 case

[illegible]

Proof of existence and uniqueness, 3×3 case

$$\begin{array}{ccc}
\begin{array}{c} \left| \begin{array}{ccc} a & d & g \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array} \right| + \left| \begin{array}{ccc} a & d & g \\ 0 & 0 & h \\ 0 & 0 & 0 \end{array} \right| + \left| \begin{array}{ccc} a & d & g \\ 0 & 0 & 0 \\ 0 & 0 & i \end{array} \right| \\
\left| \begin{array}{ccc} a & 0 & g \\ 0 & e & 0 \\ 0 & 0 & 0 \end{array} \right| + \left| \begin{array}{ccc} a & 0 & 0 \\ 0 & e & h \\ 0 & 0 & 0 \end{array} \right| + \left| \begin{array}{ccc} a & 0 & 0 \\ 0 & e & 0 \\ 0 & 0 & i \end{array} \right| \\
\left| \begin{array}{ccc} a & 0 & g \\ 0 & 0 & 0 \\ 0 & f & 0 \end{array} \right| + \left| \begin{array}{ccc} a & 0 & 0 \\ 0 & 0 & h \\ 0 & f & 0 \end{array} \right| + \left| \begin{array}{ccc} a & 0 & 0 \\ 0 & 0 & 0 \\ 0 & f & i \end{array} \right|
\end{array}
\end{array}$$

Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} +$$
$$+ \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

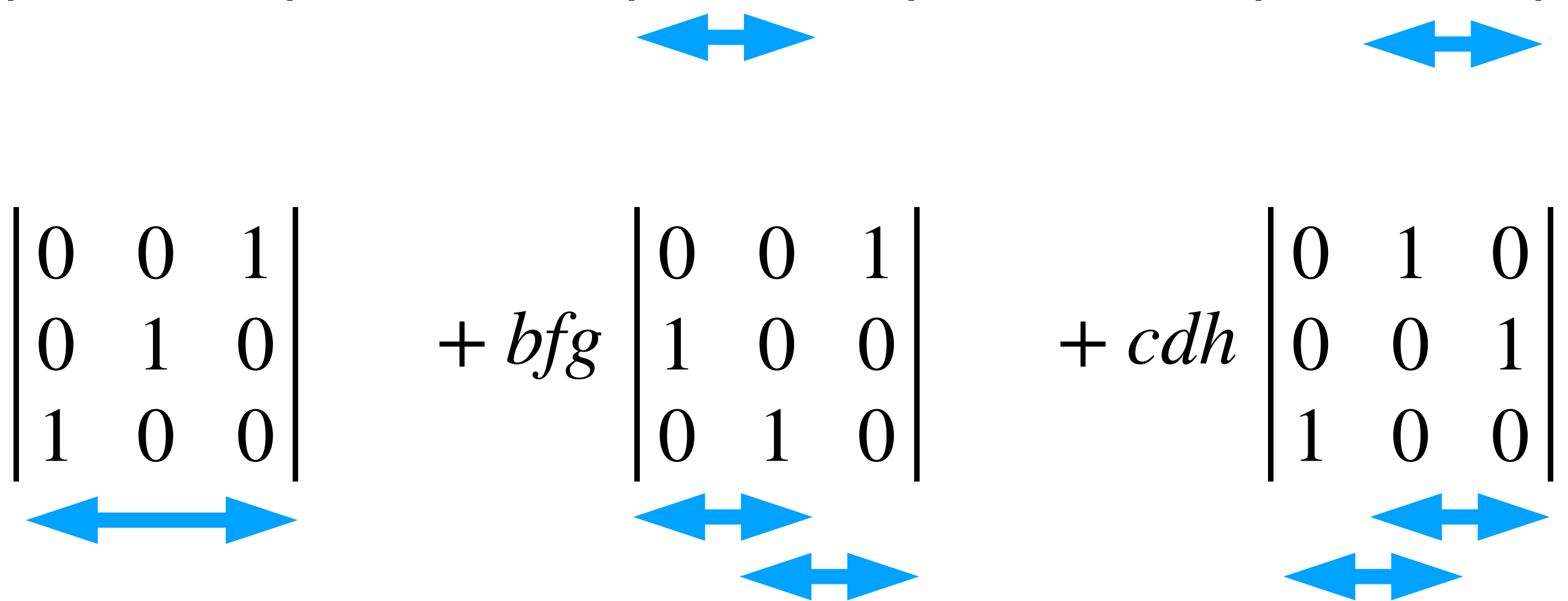
Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + bdi \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} + afh \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} +$$
$$+ ceg \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + bfg \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} + cdh \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$

Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + bdi \begin{vmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{vmatrix} + afh \begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} + \\
 + ceg \begin{vmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + bfg \begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} + cdh \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{vmatrix}$$


Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} - bdi \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} - afh \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + \\
 -ceg \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + bfg \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + cdh \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Diagram illustrating the expansion of the 3×3 determinant using the first row. The determinant is expressed as a sum of six terms, each involving a 3×3 minor matrix. Blue double-headed arrows indicate the sign changes for the minors:

- aei (positive)
- $-bdi$ (negative)
- $-afh$ (negative)
- $-ceg$ (negative)
- $+bfg$ (positive)
- $+cdh$ (positive)

Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} - bdi \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} - afh \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} +$$
$$-ceg \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + bfg \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} + cdh \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix}$$

Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \cdot 1 - bdi \cdot 1 - afh \cdot 1 + \\ -ceg \cdot 1 + bfg \cdot 1 + cdh \cdot 1$$

Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \cdot 1 - bdi \cdot 1 - afh \cdot 1 +$$

$$-ceg \cdot 1 + bfg \cdot 1 + cdh \cdot 1$$

$$= aei + bfg + cdh - afh - bdi - ceg$$

Proof of existence and uniqueness, 3×3 case

There are $3! = 6$ different orderings (permutations) of the three standard basis vectors.

$$\begin{vmatrix} a & d & g \\ b & e & h \\ c & f & i \end{vmatrix} = aei \cdot 1 - bdi \cdot 1 - afh \cdot 1 +$$

$$-ceg \cdot 1 + bfg \cdot 1 + cdh \cdot 1$$

$$= aei + bfg + cdh - afh - bdi - ceg$$

Big results about determinants

For $n \times n$ matrices A and B , $\det(AB) = \det(A) \det(B)$ and $\det(A^{-1}) = \det(A)^{-1}$ (Thm. 3.3.5)

($\det$ is a homomorphism)

$$\det A = \det A^T$$

You know the quadratic formula. What about the linear systems formula?

Why does cofactor expansion work?

Every invertible matrix is the product of elementary matrices.

But $\det(E_i)$ is easily checked, and so is $\det(AE_i) = \det(A) \det(E_i)$

This follows:

For $n \times n$ matrices A and B , $\det(AB) = \det(A) \det(B)$ and $\det(A^{-1}) = \det(A)^{-1}$ (Thm. 3.3.5)

Next up:

- HW₃ debrief today 4-5pm, Eckhart 207A
- Problem swap solutions are due tonight
- Quizzes 6 & 7 on Friday (finding a basis for a span, change of basis)
- Problem session on Friday, 3-5pm, Ryerson 177.
- HW₄ due Friday / Saturday.
- Midterm exam review on Sunday in Cobb 115, 3:00-4:30pm
- Midterm exam on Monday